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Master's Thesis

Interventions in Asymmetric Belief Systems

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Declaration of Authorship

I, Henry Zwart, declare that this thesis, entitled 'Interventions in Asymmetric Belief Systems' and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a research degree at the University of Amsterdam.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
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- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Signed:

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Abstract

Faculty of Science
Informatics Institute

Master of Science in Computational Science

Interventions in Asymmetric Belief Systems

by Henry Zwart

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Acknowledgements

Thank the people that have helped: supervisors, family, etc.

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Abbreviations

CSL **C**omputational **S**cience **L**ab

UvA **U**niversiteit **v**an **A**msterdam

Chapter 1

Terminology and notation

Define key terminology, outline notation used in following sections.

Chapter 2

Introduction

2.1 Plan

- Motivate the problem
- Contributions
- Research questions (thinking these are perhaps better left to later, in favour of contributions here)

2.2 Motivation

Issue support on climate policies in US driven by political identification and climate beliefs (Bumann, 2021; Roser-Renouf et al., 2014; Shao & Hao, 2020; Unsworth & Fielding, 2014; Ziegler, 2017).

But political identification is often not consistent with policy attitudes (Huddy et al., 2015; Iyengar et al., 2012). More often identity-driven (symbolic) than issue-driven (operational) (Mason, 2018). Also (Egan, 2020).

2.3 Research questions

Theoretical contributions:

- **RF1.1:** Extending the causal attitude network model of belief systems to: (i) support asymmetric causal effects between beliefs and attitudes, and (ii) model intervention dynamics.

Inferring belief systems:

- **RQ2.1:** To what extent are causal relations *symmetric* or *asymmetric* in models of climate change belief systems inferred from observational data?

- **RQ2.2:** How do (symmetric or asymmetric) belief systems relating to climate change vary between conservative and liberal individuals?

Intervention dynamics:

- **RQ3.1:** How do asymmetric and symmetric beliefs systems inferred from the climate attitudes dataset differ with regards to intervention strategy and effectiveness?

- **RQ3.2:** How do intervention outcome and effectiveness vary between individuals with different initial conditions, or between conservative and liberal individuals?

2.4 Contributions

1. We present a mathematical model for belief system dynamics that does not assume equilibrium and does not assume symmetric influence between cognitive aspects (Chapter 3)
2. We describe a novel parameter estimation method for fitting binary Ising models to continuous data (Chapter 4).
3. We calibrate said model to data from a recent longitudinal survey including items on beliefs and attitudes regarding climate change (Chapter 6).
4. We demonstrate, by way of the calibrated model, the existence of asymmetric influence relations between beliefs and attitudes. Furthermore we show that influence relations are not *necessarily* asymmetric, may vary in the degree of asymmetry, and can be unidirectional (Chapter 6).
5. We demonstrate that the decision to represent asymmetric relations in belief system models can change intervention dynamics, and therefore conclusions one draws regarding intervention effect and effectiveness (Chapter 6).
6. We then show that belief systems may vary significantly between individuals, by fitting the proposed model to subsets of the climate attitudes dataset comprising conservative and liberal individuals (Chapter 7).
7. Finally, we show that reasoning about the effects of interventions on *individuals* is, in general, non-trivial. How an individual responds to an intervention typically depends on their prior belief system state, including beliefs and attitudes other than the target and goal of intervention (Chapter 7).

Chapter 3

Non-equilibrium belief systems

3.1 Belief system dynamics

We will now formalise the dynamics of an individual belief system, under the following assumptions:

1. Beliefs and attitudes can be treated as discrete entities.

2. The state of a belief or attitude is Markovian with respect to the previous system configurations.

3. The probability distribution relating current and future belief states does not change over time.

The first assumption includes two key points. Firstly, that it makes sense to consider beliefs and attitudes as *entities*, or *objects*. This contrasts, for instance, alternative interpretations of attitudes as emissions from an underlying structure. (**TODO:** review CAN paper for further discussion on this matter). Secondly, that beliefs and attitudes can be considered *discrete* or *separable*. This is to say that there are no absolute physical constraints on the combinations of states which may be observed (although certain combinations may be highly unlikely). In particular, this captures the requirement that each belief or attitude can be represented using a single state variable.

Our second assumption concerns the treatment of beliefs and attitudes as Markovian.

- Harder to justify conceptually, since individuals have memory and can recall prior states beyond the previous second.

Consider a belief or attitude S_i with domain Ω_{S_i}. The state of S_i at time t depends on the previous states of other beliefs and attitudes, and we can therefore describe the trajectory of S_i as a Markov process:

$$\{s_i^t\}_{t=1}^{\infty} \quad \text{where} \quad s_i^t \in \Omega_{S_i} \tag{3.1}$$

with conditional state probability $P\big(S_i^{\tau+1} \mid \boldsymbol{S}^1, ..., \boldsymbol{S}^\tau\big) = P\big(S_i^{\tau+1} \mid \boldsymbol{S}^\tau\big)$ for $\tau \in \mathbb{N}$.

Since the state of S_i^{t+1} depends only on the previous belief system state, it follows that for any other belief or attitude S_j, the states S_i^{t+1} and S_j^{t+1} are conditionally independent given $\boldsymbol{S}^t$. We can therefore straightforwardly extend Equation 3.1 to describe the evolution of the entire belief system as a Markov process,

$$\{\boldsymbol{s}^t\}_{t=1}^{\infty} \quad \text{where } \boldsymbol{s}^t \in \Omega_{S_1} \times \dots \times \Omega_{S_n} \tag{3.2}$$

with the probability of each belief system configuration given by the conditional distribution $P\big(\boldsymbol{S}^{t+1} \mid \boldsymbol{S}^t\big)$.

Under this conceptualisation, the task of modelling a belief system thus reduces to describing how the instantaneous configuration of belief states affects the probability distribution over possible future states.

3.2 Non-equilibrium belief system model

When studying the *symmetric* Ising model it is common to assume that the model is at equilibrium, such that $P\big(\boldsymbol{S}^{t+1} = \boldsymbol{s} \mid \boldsymbol{S}^t\big) = P(\boldsymbol{S} = \boldsymbol{s})$ for all configurations $\boldsymbol{s}$ and observation times t. The probability of observing $\boldsymbol{s}$ is then time-invariant and given by the Boltzmann probability parameterised by the Hamiltonian $H(\boldsymbol{s})$ (Cardy, 1996; Christensen & Moloney, 2005):

$$P(\boldsymbol{S} = \boldsymbol{s}) = \frac{e^{-\frac{1}{k_b T} \cdot H(\boldsymbol{s})}}{\sum_{\boldsymbol{s}'} e^{-\frac{1}{k_b T} \cdot H(\boldsymbol{s}')}} \tag{3.3}$$

where k_b is the Boltzmann constant, and the temperature parameter $T \in \mathbb{R}^+$ controls stochasticity. Equation 3.3 converges to a uniform distribution over all configurations when $T \to +\infty$, and a uniform distribution over the restricted set of configurations for which $H(\boldsymbol{s}')$ is minimised when $T \to 0^+$.

For the purposes of this study, we are interested in intervention dynamics, which are inherently non-equilibrium. To see why this is the case, notice that the purpose of intervention is to change the distribution of observed states. This is true both interventions intended to change the dominant state (e.g., to promote a sustainable alternative to a typical unsustainable behaviour) or reinforce it. Any belief system model intended for studying intervention dynamics therefore cannot assume equilibrium, and cannot define model dynamics using the Boltzmann distribution.

We instead define the conditional distribution $P(\boldsymbol{S}^{t+1} = \boldsymbol{s} \mid \boldsymbol{S}^t)$ explicitly. Recall that the states of each pair of spins S_i, S_j at time t are conditionally independent random variables, given knowledge of the previous configuration $\boldsymbol{S}^t$ (Chapter 3.1). It therefore suffices for us to describe the distribution over spin states for an individual spin S_i at time $t+1$, as conditional on $\boldsymbol{S}^t$.

As in the standard Ising model formulation, we will assume that the model typically evolves in the direction of lower energy; however, rather than defining the energy at the level of the model using the Hamilonian, we consider the energy experienced by S_i as a result of its baseline activation (also known as a *local field effect*), and interactions with other spins.

The baseline activation $h_i \in \mathbb{R}$ describes the tendency for S_i to take on the value $+1$ in the absence of interactions with other spins, or, more generally, when pairwise spin interactions have a net-effect of zero. S_i tends to adopt the value $+1$ under these circumstances, iff, $h_i > 0$, and -1 iff $h_i < 0$. This tendency increases with the magnitude of h_i.

Other spins influence S_i's state through alignment or opposition relations. For a spin S_j, we define an interaction effect J_{ji} (read *'influence of j on i'*). When J_{ij} if positive or negative, S_i is influenced to adopt the state S_j^t or $-S_j^t$ respectively. In the special case when $j = i$, we refer to J_{ii} as a *self-influence* effect. Positive self-influence effects reflect the inertia of S_i, i.e., the tendency to sustain a particular belief or attitude irrespective of other beliefs and attitudes. Negative self-influence effects are not clearly interpretable. The inclusion of self-influence effects allows us to capture the timescales of different beliefs or attitudes independently.

The energy experienced by S_i for a particular spin state $s \in \{-1, +1\}$ is then the result of s in combination with the baseline activation and influence effects from all spins with edges to S_i:

$$H_i(s \mid \boldsymbol{s}^t) = h_i \cdot s + \sum_j A_{ji} J_{ji} s_j^t \cdot s \tag{3.4}$$

where $\boldsymbol{A} \in \{0,1\}^{n \times n}$ is a pairwise adjacency matrix, with $A_{ji} = 1$, iff, S_j influences S_i. Dividing the common factor of s, we can equivalently write Equation 3.4 as

$$H_i(s \mid \boldsymbol{s}^t) = s \cdot h_i^{\text{eff}}(\boldsymbol{s}^t) \tag{3.5}$$

where $h_i^{\text{eff}}(\boldsymbol{s}^t) = h_i + \sum_j A_{ji} J_{ji} s_j^t$ is the effective baseline activation experienced by S_i at time t. We then define the probability that S_i adopts the state s at time t as:

$$\begin{aligned} P\big(S_i^{t+1} = s \mid \boldsymbol{S}^t = \boldsymbol{s}\big) &= \frac{e^{\frac{1}{T} H_i(s \mid \boldsymbol{s})}}{\sum_{s' \in \{0,1\}} e^{\frac{1}{T} H_i(s' \mid \boldsymbol{s})}} \\ &= \frac{e^{\frac{1}{T} H_i(s \mid \boldsymbol{s})}}{e^{\frac{1}{T} H_i(s \mid \boldsymbol{s})} + e^{-\frac{1}{T} H_i(-s \mid \boldsymbol{s})}} \\ &= \frac{e^{\frac{1}{T} s \cdot h_i^{\text{eff}}(\boldsymbol{s})}}{e^{\frac{1}{T} s \cdot h_i^{\text{eff}}(\boldsymbol{s})} + e^{-\frac{1}{T} s \cdot h_i^{\text{eff}}(\boldsymbol{s})}} \end{aligned} \tag{3.6}$$

For $s = +1$, this reduces to the logistic function:

$$\begin{aligned} p_i^{\boldsymbol{s}} := P\big(S_i^{t+1} = +1 \mid \boldsymbol{S}^t = \boldsymbol{s}\big) &= \frac{e^{\frac{1}{T} \cdot h_i^{\text{eff}}(\boldsymbol{s})}}{e^{\frac{1}{T} \cdot h_i^{\text{eff}}(\boldsymbol{s})} + e^{-\frac{1}{T} \cdot h_i^{\text{eff}}(\boldsymbol{s})}} \\ &= \frac{1}{1 + e^{-2 \cdot \frac{1}{T} \cdot h_i^{\text{eff}}(\boldsymbol{s})}} \\ &= \text{logistic}\big(2 h_i^{\text{eff}}(\boldsymbol{s})/T\big) \end{aligned} \tag{3.7}$$

Therefore the complete conditional distribution is given by:

$$\begin{aligned} P\big(\boldsymbol{S}^{t+1} = \boldsymbol{s}^{t+1} \mid \boldsymbol{S}^t = \boldsymbol{s}^t\big) &= \prod_{i=1}^{n} P\big(S_i^{t+1} = s_i \mid \boldsymbol{S}^t = \boldsymbol{s}^t\big) \\ &= \prod_{i=1}^{n} \Big[\Big(\frac{1 + s_i}{2}\Big) p_i^{\boldsymbol{s}^t} + \Big(\frac{1 - s_i}{2}\Big)\big(1 - p_i^{\boldsymbol{s}^t}\big)\Big] \end{aligned} \tag{3.8}$$

Where the initial distribution $P(\boldsymbol{S}^0)$ is specified explicitly.

3.3 Symmetric and asymmetric belief systems

Let $\mathcal{M}$ be a belief system model defined as in the preceding section, comprising $N \in \mathbb{N}$ beliefs or attitudes, with adjacency matrix $\boldsymbol{A} \in \{0,1\}^{N \times N}$, interaction effect matrix $\boldsymbol{J} \in \mathbb{R}^{N \times N}$, and baseline activations $\boldsymbol{h} \in \mathbb{R}^N$.

When each entry in $\boldsymbol{J}$ is independent of all others, we say that $\mathcal{M}$ is an **asymmetric belief system**. The term *asymmetric* here refers to the directed relations between a pair of nodes. In an asymmetric belief system, for any pair of distinct nodes $S_i \neq S_j$, it may be the case that an influence relation exists only in one direction, or that the directed relations differ in magnitude.

In the special case where we constrain $A_{ij} = A_{ji}$ and $J_{ij} = J_{ji}$ for all $i, j \in [1, N]$, we instead say that $\mathcal{M}$ is a **symmetric belief system**. In a symmetric belief system all interactions are directionally-equivalent. This can be considered a simple extension of the symmetric Ising model with (i) self-interaction terms and (ii) temporal dynamics, thus the symmetric belief system model tends toward an equilibrium steady state.

3.4 Simulation via Glauber dynamics

It is straightforward to draw samples from a belief system model using Glauber dynamics, given the conditional probability distribution defined in Equation 3.8.

Given an initial state $\boldsymbol{s}^0 \in \{-1, +1\}^N$, we sample a sequence of $T \in \mathbb{N}$ subsequent configurations:

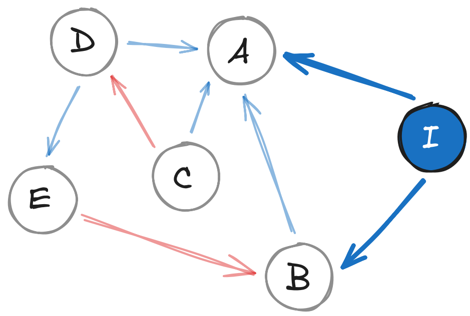
$$\{\boldsymbol{s}^t\}_{t=0}^{T}, \quad \text{where each } \boldsymbol{s}^{t+1} \sim P\big(\boldsymbol{S}^{t+1} \mid \boldsymbol{S}^t\big) \tag{3.9}$$

Each belief or attitude has the opportunity to update during every time interval. This update routine is referred to as *synchronous* Glauber dynamics, contrasting *asynchronous* Glauber dynamics, in which only one spin can update during a given interval.

3.5 Modelling interventions

We now outline our approach to modelling interventions in the asymmetric belief system model. In this study we consider interventions which affect the *state* of a belief system. Notably, this excludes interventions intended to influence the structure of a belief system, for instance, by changing the existence of effect size of influence relations between beliefs and attitudes.

Let $\mathcal{M}$ be a belief system model with parameters $\langle \boldsymbol{A}, \boldsymbol{J}, \boldsymbol{h} \rangle$. We can consider an intervention as an auxiliary node, I, in the belief system network, with state fixed at a particular value and outgoing edges toward a subset of beliefs and attitudes:



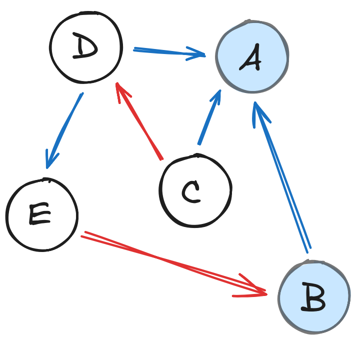
The intervention node I exerts influence on the belief system nodes A and B. Let us consider the effective baseline activation at node A and time $t+1$, as defined in Equation 3.4, given a previous configuration $\boldsymbol{s}^t$:

$$h_A^{\text{eff}}(\boldsymbol{s}^t) = h_A + \sum_{\text{node } j} A_{jA} J_{jA} s_j^t \tag{3.10}$$

Since the state of I is fixed (in this case, at $+1$), the contribution of the intervention node to A's effective activation baseline is constant. We can then re-write Equation 3.10, interpreting the intervention effect as an adjustment to the baseline activation h_A:

$$h_A^{\text{eff}}(\boldsymbol{s}^t) = (h_A + J_{IA}) + \sum_{\text{node } j \neq I} A_{jA} J_{jA} s_j^t \tag{3.11}$$

It follows then that we can model interventions more simply as an adjustment to the baseline activation of certain beliefs and attitudes:



Let us define this more formally. For a belief system model $\mathcal{M}$ with $N \in \mathbb{N}$ nodes, and parameters $\langle \boldsymbol{A}, \boldsymbol{J}, \boldsymbol{h} \rangle$, an *intervention*, is the function $\varphi_{\mathcal{M}} : \delta_h \mapsto \mathcal{M}'$, where $\delta_h \in \mathbb{R}^N$ is a vector of offsets to the baseline activations, and the model $\mathcal{M}'$ has parameters $\langle \boldsymbol{A}, \boldsymbol{J}, \boldsymbol{h}' \rangle$, where

$$\boldsymbol{h}' = \boldsymbol{h} + \delta_h \tag{3.12}$$

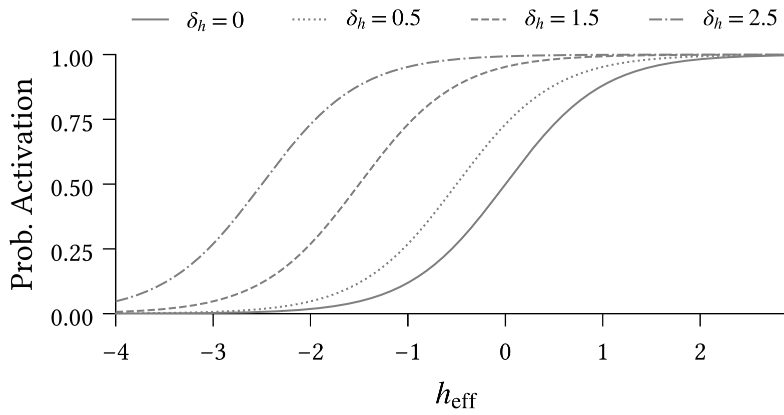


Figure 3.3: Impact of different intervention strengths ($\delta_h \in \{0, 0.5, 1.5, 2.5\}$) on activation probability for the intervention spin. Activation probability is calculated using Equation 3.7.

Figure 3.3 illustrates how the probability of activation for the intervention spin (Equation 3.7) changes for different intervention scenarios. Recall that a spin's activation probability is a function of it's effective local field. Since interventions constitute offsets to the effective local field, they are reflected as horizontal shifts in the logistic curve.

The solid line denotes the null scenario in which no intervention is applied. Consider two individuals, positioned at different locations on this base curve:

1. $h_{\text{eff}} = 0$: Ambivalent disposition toward the intervention spin, and

2. $h_{\text{eff}} = -2$: Strong negative disposition toward the intervention spin.

To understand the impacts of intervention strength on each individual, we draw a vertical line upward from the solid curve at each location, and read off the activation probability at each intersection with the other intervention curves. Under the null scenario the first individual exhibits an equal tendency toward each state, while the second is almost certain to adopt the state -1.

Notice that interventions are not experienced equivalently by the two individuals. For a weak intervention ($\delta_h = 0.5$), the first individual sees a substantial jump in activation probability, while the corresponding increase for the second individual is negligible. The stakes are reversed under a strong intervention ($\delta_h = 2.5$).

We can understand this difference in effect both conceptually and analytically. From a conceptual perspective, we can explain the large shift of the first individual under a weak intervention as resulting from their prior indifference, making them maximally susceptible to intervention. Prior to intervention, the second individual has a strong negative stance on this belief/attitude; weak interventions are insufficient to change their view. On the other hand, the second individual's negative stance provides more space for their attitude or belief to shift under a successful intervention. Compare this to the first individual, who is shifting from a place of indifference.

TODO: Quantitative explanation:

- Take derivative of the difference in activation probabilities between intervention and null scenario; solve for derivative equal to zero. Corresponds to the unique maximum since for $h_{\text{eff}} \to \pm\infty$ the second derivative is strictly positive.
- We find that maximum is at $h_{\text{eff}} = -\frac{\delta_h}{2}$.
- i.e., for a given intervention strength, the maximum change in activation probability occurs when the intervention causes h_{eff} to increase to same magnitude on the other side of zero.
- Individuals below this point require a stronger intervention to move the same amount.
- Individuals above this point experience ceiling effect. A smaller intervention could achieve almost the same effect.

Chapter 4

Methods

4.1 Counterfactual interventions

Individuals have different belief systems. Influence relations between beliefs and attitudes reflect heterogeneity in individual experiences and perception. Often these relations can themselves be considered beliefs — whether or not support for climate change mitigation leads to support for any specific policy depends in-part on an individual's belief regarding the policy's relevance. An individual's baseline activations, in turn, reflect a culmination of factors which determine their tendency toward certain beliefs or attitudes.

Our approach to parameter estimation, however, fundamentally assumes that the inferred belief system is shared by all individuals in the dataset from which the model is estimated. We note that this is not strictly a limitation of the parameter estimation method, nor of the model itself. Rather this assumption is imposed by data limitations. Fitting individual models requires substantial data from each individual, which is not available in the climate attitudes dataset. Moreover, even with substantial observational data, we are unlikely to observe all state transitions necessary to understand the *potential* dynamics of an individual system, owing to the relatively slow-moving nature of beliefs and attitudes.

For any given individual, the inferred models may thus not accurately reflect the stability of belief system states. A configuration which is stable in a given individual's own belief system may be considered unstable in the shared model. When simulating interventions, we take the most recent measurement for each individual as their initial state. However, due to this perceived instability, we expect to observe natural dynamics away from this initial state, even in null-intervention scenarios.

We thus consider intervention effects as relative to the null-intervention counterfactual baseline rather than relative to the initial state. For each intervention we run both intervention and null ($\delta_h = 0$) scenarios with identical random number generation settings, and define the *intervention effect* as the difference between the observed outcomes.

4.2 Binarisation

Prior to model fitting, we binarise all variables, mapping data values to the domain $\{-1, +1\}$. Before binarising, we shift each variable such that the *survey midpoint* (e.g., '3' on a 5-point Likert scale) aligns with 0, and rescale each variable such that the minimum and maximum possible values map to -1 and $+1$ respectively. For variables with no well-defined midpoint, such as those with an even number of possible responses, we shift such that the minimal and maximal values are at equal distance from 0.

We use a smooth binarisation process to robustly handle data points which are zero, or close to zero. This involves perturbing each data value $x \in \mathbb{R}$ by an independent noise term $\varepsilon \sim \mathcal{N}(0, \sigma)$ before thresholding. Figure 4.1 illustrates this process for a negative value of x and given choice of σ.

Observe that x is mapped to $+1$ if, and only if, ε is sufficiently large, such that $x + \varepsilon > 0$ (region A), or equivalently when $\varepsilon < x$ (region B). The probability that x is mapped to $+1$ is then

$$P(x \mapsto +1) = \Phi\left(\frac{x}{\sigma}\right) \tag{4.1}$$

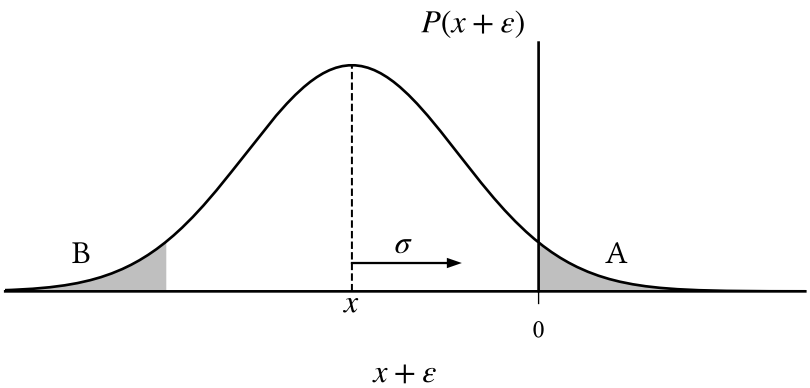


Figure 4.1: $x \in \mathbb{R}$ is smoothly binarised to $\{-1, +1\}$ by thresholding $x' = (x + \varepsilon)$ where $(x + \varepsilon) \sim \mathcal{N}(x, \sigma)$. Negative values are mapped to $+1$ with probability $P(x' > 0) = \mathrm{A} = \mathrm{B} = P(\varepsilon < x)$, which increases with σ and $|x|^{-1}$.

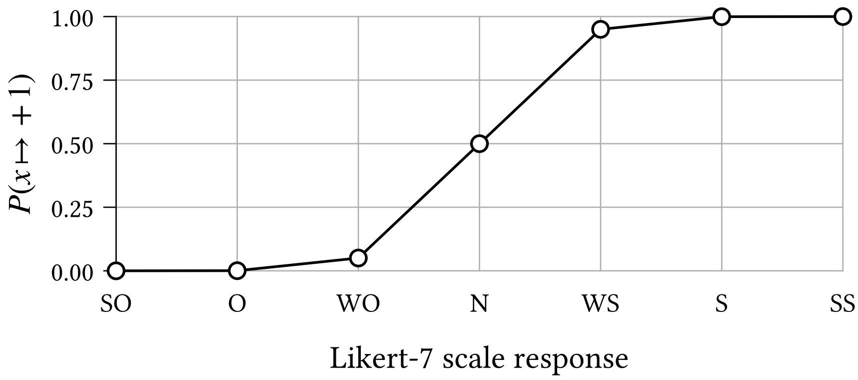


Figure 4.2: The probability of binarisation to +1 for each possible response to a Likert-7 scale survey question[1], given $\sigma \approx 0.2$.

where Φ is the standardised normal cumulative distribution function. This probability is small when x has large magnitude, or when σ is small. For the purposes of our experiments, we choose $\sigma = 0.203$ such that a 'weakly oppose' response to a 7-point Likert scale[1] (value $1/3$) is mapped to $+1$ with probability 0.05.

4.3 Parameter estimation

For the purposes of the experiments in the following sections, we perform parameter estimation to calibrate the belief system model described in Chapter 3.2 to the climate attitudes dataset (Chapter 9). We first outline the context of the parameter estimation problem, and then present our approach. We omit derivation details in this section; however, the curious reader can find these in Appendix A (**TODO**).

Consider a population of $M \in \mathbb{N}$ individuals with a shared belief system $\mathcal{M}$ comprising $N \in \mathbb{N}$ beliefs or attitudes with pre-defined adjacency matrix $\boldsymbol{A} \in \{0,1\}^{N \times N}$, for which the parameters (i.e., the baseline activations and interaction effects) are unknown. Suppose that for each individual $m \in [1, M]$ we have observed a series of measurements, reflecting that individual's belief system state at each of $t \in [1, T]$ uniformly spaced timesteps:

$$\left\{\boldsymbol{x}^t_{(m)}\right\}_{t=1}^{T}, \quad \text{where each } \boldsymbol{x}^t_{(m)} \in \mathbb{R}^N \tag{4.2}$$

The measurements need not be binary, but are assumed to be real. The collection of measurements across all individuals forms a dataset D, which is the particular observed value of the random variable $\mathcal{D}$ representing possible datasets.

[1]The oppose/support 7-point Likert scale has possible responses: strongly oppose, oppose, weakly oppose, neutral, weakly support, support, strongly support.

While maximum likelihood estimation (MLE) is often used to infer parameters for similar models (Lee et al., 2015; Nguyen et al., 2017), we should be cautious, for two reasons, of applying this method naïvely in the present study.

Firstly, the model defined in Equation 3.8 assumes binary spin states $s \in \{-1, +1\}$. The climate attitudes dataset does not satisfy this assumption and must be binarised for MLE to be applicable. However, we have no guarantee that the parameterisation inferred for any specific binarisation is a reasonable explanation for the non-binarised dataset, nor for any other possible binarisation in the case where a probabilistic binarisation scheme is used (such as the one described in Chapter 4.2).

Secondly, maximum likelihood estimation is prone to overfitting when the number of model parameters is similar to the number of observations (Epskamp, Borsboom, et al., 2018). While we consider only a small number of beliefs and attitudes in this study ($N \in \{7, 8\}$), the number of parameters p grows rapidly, with $p \in O(N^2)$ for both the symmetric and asymmetric model variants. In combination with sampling error, we expect the number of parameters to negatively impact the robustness of the inferred parameters — given an alternative dataset of the same size, we would expect significant variation in parameters.

We mitigate these respective issues in our parameter estimation scheme by (i) marginalising over the set of possible binarisations of D, and (ii) applying regularisation based on parameter magnitudes. We choose the parameterisation $\boldsymbol{\theta}^* \in \mathbb{R}^p$ which satisfies the optimisation problem:

$$\boldsymbol{\theta}^* = \operatorname*{argmax}_{\hat{\boldsymbol{\theta}} \in \mathbb{R}^p} f\left(D; \hat{\boldsymbol{\theta}}\right), \quad \text{where} \quad f\left(D; \hat{\boldsymbol{\theta}}\right) := \mathcal{L}_D\left(\hat{\boldsymbol{\theta}}\right) - \lambda \sum_{\theta \in \hat{\boldsymbol{\theta}}} \sqrt{\theta^2 + \varepsilon} \tag{4.3}$$

The first term in the objective function f, $\mathcal{L}_D\left(\hat{\boldsymbol{\theta}}\right)$, denotes the *expected* log-likelihood given a parameterisation $\hat{\boldsymbol{\theta}}$, over possible binarisations of D:

$$\mathcal{L}_D\left(\hat{\boldsymbol{\theta}}\right) = \mathbb{E}\left[L_{D_B}\left(\hat{\boldsymbol{\theta}}\right) \mid \mathcal{D} = D\right] = \sum_{D_B} P(\mathrm{Bin}(\mathcal{D}) = D_B \mid \mathcal{D} = D) \cdot L_{D_B}\left(\hat{\boldsymbol{\theta}}\right) \tag{4.4}$$

where $L_{D_B}\left(\hat{\boldsymbol{\theta}}\right)$ is the log-likelihood of a specific binarisation D_B. We will defer explicitly defining $L_{D_B}\left(\hat{\boldsymbol{\theta}}\right)$ for now, but will return to this point shortly in Equation 4.5.

TODO: Intuition for the effect of using the expectation.

- Inferred model reflects ambiguity in states close to zero.
- The likelihood contribution of each data value is the expectation over possible spin states.

- Values close to zero contribute a comparable weighting from each binary state.
- When the likelihood given one binary state would be significantly higher than for the other, this is downweighted to account for the probability that the value does not obtain this state.
- Therefore the expectation prevents overconfident optimisation toward any specific binarisation — the inferred model is required to explain the binarised dataset *in expectation*.
- In other words, if a data value may be binarised to -1 or $+1$ with equal probability, then both cases should be reasonably explained by the parameterisation.

The second term is a smooth variant of L1 regularisation (Tibshirani, 1996). This has the effect of penalising nonzero parameters which do not meaningfully contribute to the expected likelihood. Parameters instead must reflect relationships which are sufficiently prevalent in the data, reducing the robustness issues resulting from sampling error and a large number of parameters. The hyperparameter $\lambda \in \mathbb{R}^+$ controls regularisation strength, with higher values resulting in sparser models. As $\varepsilon \to 0^+$ the second term converges to the standard formulation of L1 regularisation. Unlike the standard formulation, when $\varepsilon > 0$ the first-derivative of this term exists, so is amenible to gradient-based optimisation. We discuss the choice of values for λ and ε in Chapter 4.3.1.

Let $D_B \sim \mathrm{Bin}(D)$ be a possible binarisation of D, and $\boldsymbol{\theta}$ a parameterisation for the (asymmetric or symmetric) belief system model, which is decomposable into the model parameters $\boldsymbol{\theta} = \langle \boldsymbol{J}, \boldsymbol{h} \rangle$. The log-likelihood of D_B given $\boldsymbol{\theta}$ is:

$$\begin{aligned} L_{D_B}(\boldsymbol{\theta}) &= \log P(D_B \mid \boldsymbol{\theta}) \\ &= \sum_{m=1}^{M} \sum_{t=1}^{T-1} \log s_{i,(m)}^{t+1} \cdot h_i^{\mathrm{eff}}\Big(\boldsymbol{s}_{(m)}^t\Big) - \log\Big(2 \cosh h_i^{\mathrm{eff}}\Big(\boldsymbol{s}_{(m)}^t\Big)\Big) \end{aligned} \tag{4.5}$$

The derivation of Equation 4.5 is analogous to that of the non-equilibrium Ising model log-likelihood (Nguyen et al., 2017). Combining Equation 4.5 and Equation 4.4, we obtain an explicit expression for the expected likelihood:

$$\mathcal{L}_D(\boldsymbol{\theta}) = \sum_{\boldsymbol{s}} \left(\sum_{m=1}^{M} \sum_{t=1}^{T-1} P\Big(\boldsymbol{S}_{(m)}^t = \boldsymbol{s} \mid D\Big) \cdot \mathbb{E}\Big[\boldsymbol{S}_{(m)}^{t+1} \mid D\Big]^T h^{\mathrm{eff}}(\boldsymbol{s}) \right) - Z(D; \boldsymbol{\theta}) \tag{4.6}$$

where $\mathbb{E}\Big[\boldsymbol{S}_{(m)}^{t+1} \mid D\Big]$ is the expected binary configuration for the observation $\boldsymbol{x}_{(m)}^{t+1}$, the vector $h^{\mathrm{eff}}(\boldsymbol{s})$ contains the effective baseline activation for each spin given the previous state $\boldsymbol{s}$, and Z is the expected log partition function, summed across observations:

$$Z(D;\boldsymbol{\theta}) = \sum_{\boldsymbol{s}} \sum_{\substack{m \le M \\ t<T}} P\Big(\boldsymbol{S}^t_{(m)} = \boldsymbol{s} \mid D\Big) \sum_{i=1}^{N} \log\big(2\cosh h_i^{\text{eff}}(\boldsymbol{s})\big) \tag{4.7}$$

Per the optimisation problem defined in Equation 4.3, we choose the parameterisation $\boldsymbol{\theta}^*$ which maximises the regularised expected log-likelihood, which occurs when $\frac{\partial f}{\partial \theta} = 0$ for every parameter $\theta \in \boldsymbol{\theta}^*$. The partial derivative with respect to a given baseline activation parameter h_i, for $i \in [1, N]$, is given by:

$$\begin{aligned}\frac{\partial}{\partial h_i} f(D;\boldsymbol{\theta}) = &\sum_{\boldsymbol{s}} \sum_{\substack{m \le M \\ t<T}} P\Big(\boldsymbol{S}^t_{(m)} = \boldsymbol{s} \mid D\Big)\Big[\mathbb{E}\Big[S^{t+1}_{i,(m)} \mid D\Big] - \tanh\big(h_i^{\text{eff}}(\boldsymbol{s})\big)\Big] \\ &- \frac{\lambda h_i}{\sqrt{h_i^2 + \varepsilon}}\end{aligned} \tag{4.8}$$

Note that the expressions for $\mathcal{L}_D(\boldsymbol{\theta})$ and $\frac{\partial}{\partial h_i} f(D;\boldsymbol{\theta})$ are applicable to both the asymmetric and symmetric belief system models. This property does not hold for the partial derivatives with respect to interaction effects. In the asymmetric model, for $i, j \in [1, N]$, this is derived analogously to Equation 4.8 as:

$$\begin{aligned}\frac{\partial}{\partial J_{ji}} f(D;\boldsymbol{\theta}) = &\sum_{\boldsymbol{s}} \sum_{\substack{m \le M \\ t<T}} P\Big(\boldsymbol{S}^t_{(m)} = \boldsymbol{s} \mid D\Big) A_{ji} s_j \Big[\mathbb{E}\Big[S^{t+1}_{i,(m)} \mid D\Big] - \tanh\big(h_i^{\text{eff}}(\boldsymbol{s})\big)\Big] \\ &- \frac{\lambda J_{ji}}{\sqrt{J_{ji}^2 + \varepsilon}}\end{aligned} \tag{4.9}$$

In the symmetric belief system model, since we require that $\boldsymbol{J} = \boldsymbol{J}^T$, we estimate a single interaction parameter $J_{ji} = J_{ij} = \beta_{\{i,j\}}$ for each pair of spins S_i, S_j. Each pair thus contributes twice to the partial derivative, once for each direction:

$$\begin{aligned}\frac{\partial}{\partial \beta_{\{i,j\}}} f(D;\boldsymbol{\theta}) = &\sum_{\boldsymbol{s}} \sum_{\substack{m \le M \\ t<T}} P\Big(\boldsymbol{S}^t_{(m)} = \boldsymbol{s} \mid D\Big) \sum_{(i',j') \in \{i,j\}} A_{j'i'} s_{j'} \Big[\mathbb{E}\Big[S^{t+1}_{i',(m)} \mid D\Big] - \tanh\big(h_{i'}^{\text{eff}}(\boldsymbol{s})\big)\Big] \\ &- \frac{\lambda \beta_{\{i,j\}}}{\sqrt{\beta_{\{i,j\}}^2 + \varepsilon}}\end{aligned} \tag{4.10}$$

4.3.1 Hyperparameters

The optimisation problem in Equation 4.3 has two hyperparameters which must be specified prior to parameter estimation: the regularisation strength $\lambda \in \mathbb{R}^+$, and the smoothing parameter $\varepsilon \in \mathbb{R}^+$. The values for both are summarised in Table 4.1.

Parameter	Model type	Dataset	Value
λ	Symmetric	Full	1.4×10^{-2}
	Asymmetric	Full	1.2×10^{-2}
		Conservative	3.4×10^{-2}
		Liberal	4.3×10^{-2}
ε	All	All	10^{-8}

Table 4.1: Values for regularisation strength (λ) and smoothing (ε) hyperparameters. Regularisation strength model- and dataset-specific; *Conservative* and *Liberal* refer to subsets of the climate attitudes dataset comprising individuals with the specified ideology (Chapter 9).

We use the Extended Bayesian Information Criterion (EBIC) (Chen & Chen, 2008) to select λ, as recommended by Epskamp, Borsboom, et al. (2018):

$$\mathrm{EBIC}(\boldsymbol{\theta}^*) = k \cdot [\ln(M(T-1)) + 2\gamma \ln(p)] - 2\mathcal{L}_D(\boldsymbol{\theta}^*) \tag{4.11}$$

The variable k denotes the number of nonzero parameters in the optimised model, where a parameter θ is considered nonzero if $|\theta| > 10^{-2}$. We take $\gamma = 0.25$, which has been shown to work well in general for network inference in the inverse Ising problem (Barber & Drton, 2015).

The EBIC is an evaluative criterion for model selection, which balances maximisation of the log-likelihood with model complexity, as measured by the number of parameters. Compared to the original Bayesian Information Criterion, EBIC tends to be more conservative when the number of parameters and observatons are comparable (Foygel & Drton, 2010). We select λ which minimises the EBIC for the symmetric and asymmetric models (independently), the results of which are displayed in Figure 4.3.

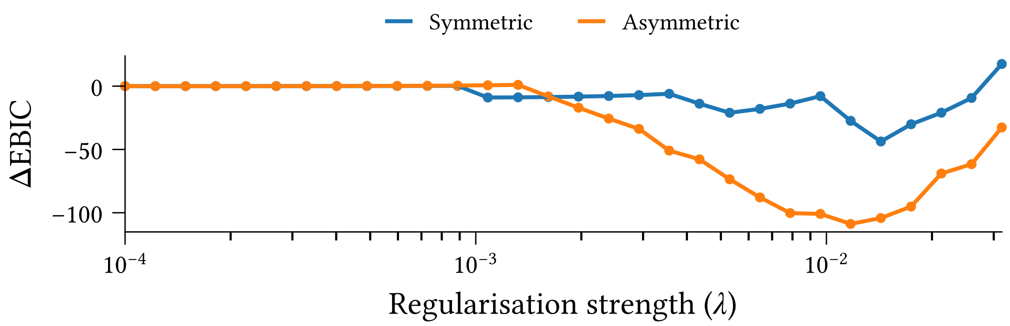


Figure 4.3: Effect of regularisation strength λ on Extended Bayesian Information Criterion for symmetric and asymmetric belief system models optimised to the climate attitudes dataset using Equation 4.3. The vertical axis measures the difference in EBIC compared to the no-regularisation case ($\lambda = 0$).

We consider parameters with magnitude less than 10^{-2} as 'effectively zero', and take $\varepsilon = 10^{-8}$ such that $\sqrt{\varepsilon}$ is significantly smaller than this threshold. This choice of ε ensures that the smooth regularisation remains a good approximation to L1 regularisation, i.e.,

$$\sqrt{\theta^2 + \varepsilon} \approx |\theta| \tag{4.12}$$

4.3.2 Implementation details

We solve the parameter estimation problem using the Scipy 1.17.1 implementation of the quasi-Newton BFGS optimisation algorithm (Nocedal & Wright, 2006, p. 136; Virtanen et al., 2020). We use the analytic jacobian comprising the partial derivatives stated above. We take $\boldsymbol{\theta} = \mathbf{0}$ as the initial guess.

To ensure numerical stability irrespective of dataset size, we rescale the expected log-likelihood contribution to the objective function and partial derivatives, dividing by the number of observed observations (time intervals) in the dataset. For $M \in \mathbb{N}$ individuals and $T \in \mathbb{N}$ timesteps this amounts to a scale factor of $\frac{1}{M(T-1)}$.

Chapter 5

Model Calibration

Before proceeding, let's take a moment to calibrate and evaluate the models which will be used in the upcoming experiments. Using the parameter estimation method outlined in the previous chapter, we calibrate the symmetric and asymmetric belief system models to the climate beliefs dataset (see Chapter 9). We will evaluate the calibrated models on both structural accuracy ('how accurate are the parameter estimates?') and predictive capacity ('how well do the models explain the data?').

Figure 5.1 shows the interaction effect matrix, $\boldsymbol{J}$, for each model. We only display the upper triangular portion of the symmetric model's matrix, since each pair of spins in

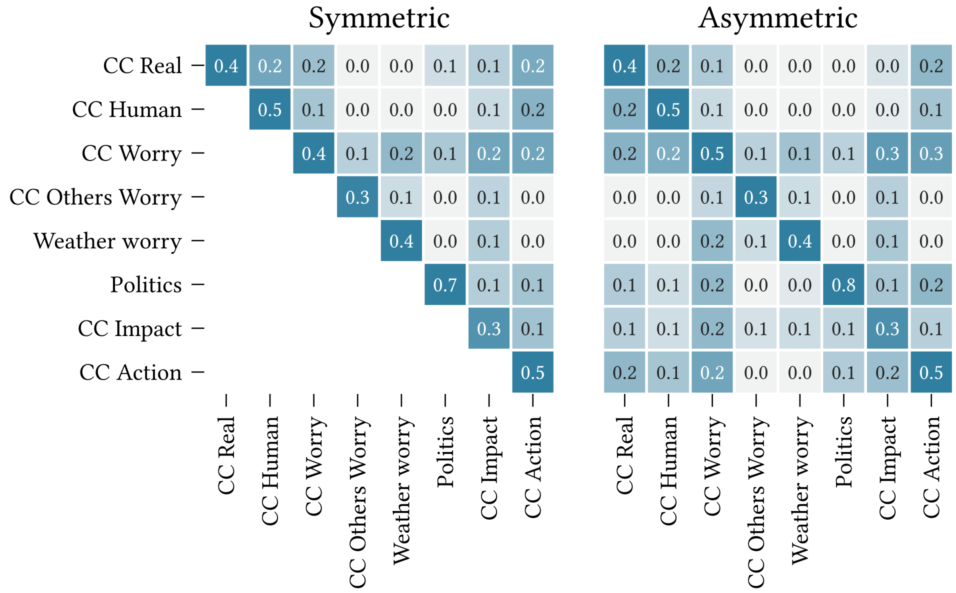


Figure 5.1: Interaction effect matrices ($\boldsymbol{J}$) for the symmetric and asymmetric belief system model variants, calibrated to the climate beliefs dataset (Chapter 9) using the parameter estimation method in Chapter 4.3.

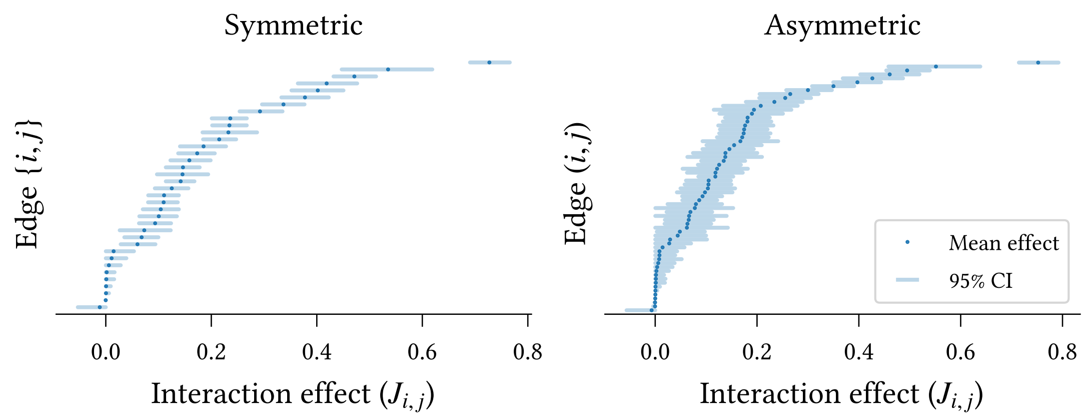


Figure 5.2: Interaction effect parameter accuracy for symmetric and asymmetric models calibrated to the climate beliefs dataset (Chapter 9). Mean parameter values are shown in increasing order. 95% confidence intervals are calculated using the percentile method over models calibrated to bootstrapped datasets (500 repeats).

this model has a single, bidirectional relation. Both models feature a dominant diagonal, indicating that most variables are slow-moving with respect to the modelled timescale[2] (they are 'sticky'). This is particularly true for `Politics`, and less so for `CC Others Worry` and `CC Impact`.

As recommended in Epskamp, Borsboom, et al. (2018), we evaluate interaction parameter accuracy by examining the bootstrapped confidence intervals around each parameter estimate. We use bootstrapping to estimate the uncertainty in our parameter estimates due to sampling error. Let $\boldsymbol{D} \in \mathbb{R}^{M \times T \times N}$ be the complete dataset, where M, T, and N denote the number of participants, observations, and spins respectively[3]. We construct 500 bootstrapped datasets by sampling rows (participants) with replacement from $\boldsymbol{D}$, such that each bootstrapped dataset has the same shape as the complete dataset.

For each bootstrapped dataset $\boldsymbol{D}_{(i)}$ we calibrate a model $\mathcal{M}_{(i)}$ with parameters

$$\langle \boldsymbol{A}, \boldsymbol{J}_{(i)}, \boldsymbol{h}_{(i)} \rangle =: \boldsymbol{\theta}^*_{(i)} \in \mathbb{R}^p \tag{5.1}$$

where $p \in \mathbb{N}$ is the number of model parameters, and we take the adjacency matrix $\boldsymbol{A}$ as fully-connected, permitting influence relations to be inferred between each pair of beliefs or attitudes.

Figure 5.2 shows the estimated interaction effects in increasing order for each model. The confidence intervals are calculated using the percentile method (**CITE**) across

[2]The models' timescales are set by the duration between survey responses, in this case approximately six months (Chapter 9).

[3]In the climate beliefs dataset we have $M = 1693$, $T = 2$, and $N = 8$

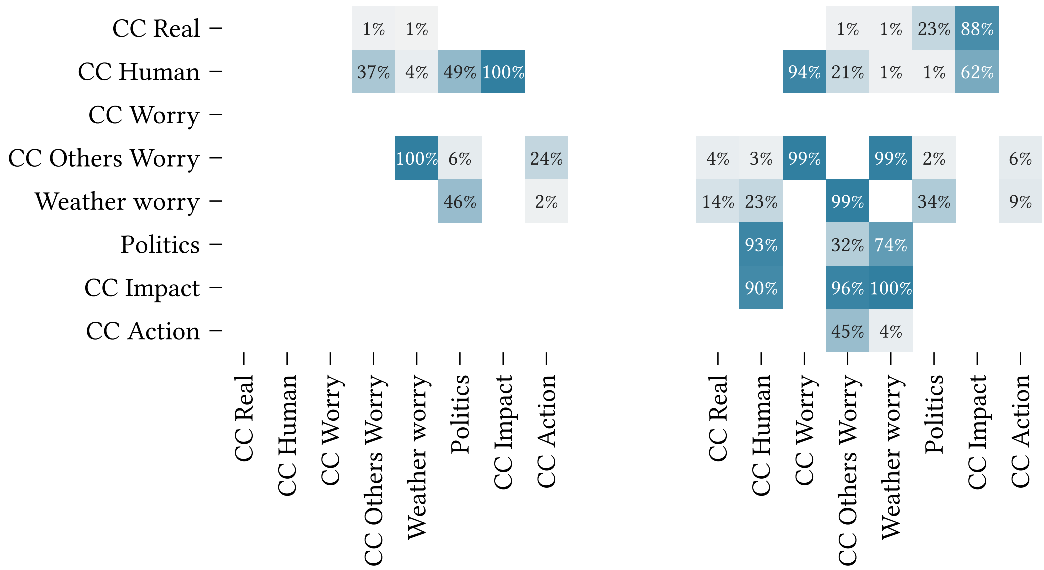


Figure 5.3: Edge selection probability for models calibrated to bootstrapped datasets sampled with replacement from the climate beliefs dataset (500 repeats). Edges are 'selected' if they survive regularisation with magnitude at least 0.01.

the bootstrapped models' parameters. Most parameters display small 95% confidence intervals, indicating an accurate fit for both models. On average, the symmetric model exhibits smaller confidence intervals than the asymmetric model ($0.065 < 0.085$), which is expected given the larger number of parameters in the asymmetric model. In both models the self-interaction effect on `Politics` is significantly larger than all other parameters, as evidenced by the non-overlapping confidence intervals, supporting our earlier observation regarding Figure 5.1. Moreover, the remaining self-interaction effects are, in-general, significantly larger than the pairwise effects, with the exception of `CC Impact` in the symmetric model, and `CC Impact`, `CC Worry Others` in the asymmetric model.

Due to the use of regularisation in model calibration we must be careful not to draw conclusions regarding edge *existence* from this figure (Epskamp, Borsboom, et al., 2018). Instead, we may consider the proportion of bootstrapped models for which each edge is nonzero, shown in Figure 5.3. Relations which are selected in all models are not shown.

All edges excluded from the model calibrated on the full dataset (Figure 5.1) have low selection probability in the bootstrapped datasets ($< 50\%$). The relation `CC Real` $\rightarrow$ `Politics` is excluded from the complete asymmetric model but included (bidirectionally) in the symmetric model. This is reflected in the corresponding selection probabilities—this edge is selected in 100% of bootstrapped symmetric models but only 23% of asymmetric models. Edges which *are* selected in the complete models generally have high selection probability. The notable exception in the asymmetric case, `CC Human` $\rightarrow$ `CC Impact`, has a relatively small effect size ($J_{i,j} \approx 0.04$).

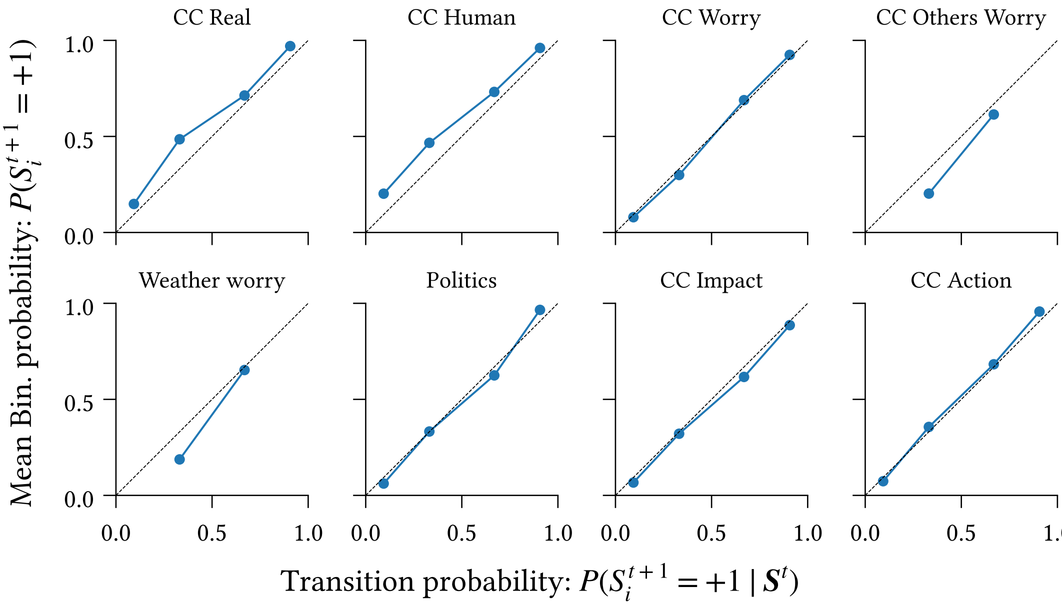


Figure 5.4: Reliability of transitions predicted by the model measured as the agreement between conditional probabilities under the model, $P\big(S_i^{t+1} = +1 \mid \boldsymbol{S}^t\big)$, and expected proportion of $+1$ values after binarisation, calculated as the average transition probability for survey participants in a given conditional probability bin.

We now evaluate the models' predictive capacities[4]. First, we examine the reliability of predicted transitions. Figure 5.4 shows the average binarisation probability for each spin (i.e., the probability that the corresponding observation in the second wave of the dataset is binarised to $+1$), binned by transition probability. Bins with no observations are not shown (e.g., extreme values for `CC Others Worry`).

The two measurements are strongly correlated for most variables. The higher variation for `CC Real` and `CC Human` likely reflects the observed heavy skew in these variables toward larger values (see Figure 9.9 in Chapter 9). We observe no high/low probabilities for either `CC Others Worry` or `Weather Worry`, on account of the limited influence of other spins on these variables, as seen in the corresponding columns of Figure 5.1.

Next, we compare the calibrated model against a null model, in which we permit only baseline activation (h_i) and self-influence ($J_{i,i}$) parameters. We measure the relative entropy of the binarised data from the second wave of the dataset, with respect to the probability distributions induced by each model. For each participant and spin, we calculate the relative entropy as

$$D(P \parallel Q) := H_C(P, Q) - H(P) \tag{5.2}$$

[4]While we only show results for the asymmetric model here, the corresponding figures for the symmetric model are substantially similar, and all conclusions drawn regarding the asymmetric model apply also to the symmetric one.

Where $P := P\big(\mathrm{Bin}\big(X_{i,(m)}^{t+1}\big)\big)$ is the distribution over binarisations of the observation in the first timestep, and $Q := P_{\mathcal{M}}\big(S_{i,(m)}^{t+1}\big)$ is the distribution over subsequent states according to the model, marginalising over binarisation of the previous observation,

$$Q := P_{\mathcal{M}}\left(S_{i,(m)}^{t+1}\right) = P_{\mathcal{M}}\left(S_{i,(m)}^{t+1} \mid \mathrm{Bin}\left(X_{i,(m)}^{t}\right)\right) \cdot P\left(\mathrm{Bin}\left(X_{i,(m)}^{t}\right)\right) \tag{5.3}$$

The functions H and H_C denote the entropy and cross-entropy, respectively. The entropy term quantifies the uncertainty in the binarisation process, and the cross-entropy measures the 'expected surprise' when comparing predictions generated from the model with the true binarised dataset. Formally, these are defined as follows:

$$H(P) = -\sum_{s\in\pm 1} P(s)\log_2 P(s) \tag{5.4}$$

$$H_C(P,Q) = -\sum_{s\in\pm 1} P(s)\log_2 Q(s) \tag{5.5}$$

The relative entropy, $D(P \parallel Q)$, is small when either (i) the model accurately predicts the observed binarised value such that the cross-entropy is low[5], or (ii) the binarisation process is very uncertain. It follows that the relative entropy is high in cases where the model fails to accurately predict the next state, despite the binarisation process being fairly deterministic.

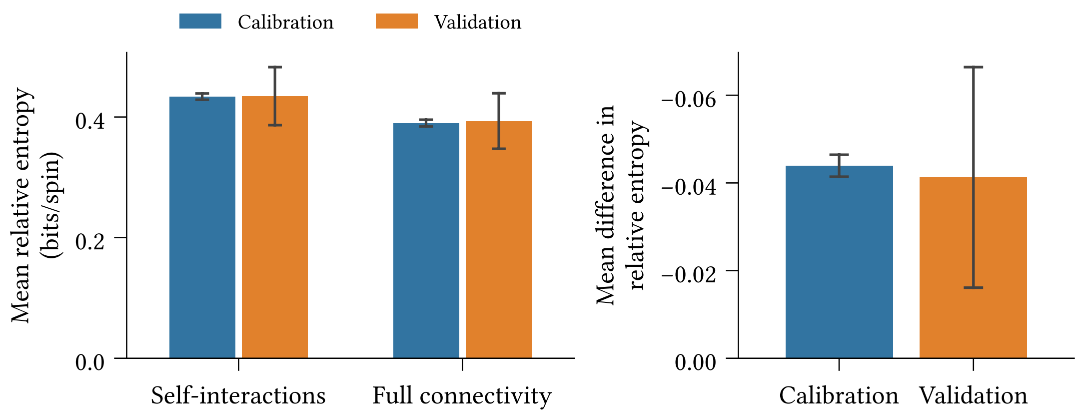


Figure 5.5: *(Left)* Mean absolute relative entropy of the binarised dataset with respect to the calibrated model, for the full-connected and null (only self-interactions) models, using 10-fold cross-validation. Average calculated across spins and survey participants. *(Right)* Mean difference in relative entropy. Confidence intervals display two standard deviations around the mean value.

[5]The relative entropy is strictly non-negative (**CITE**), so the binarisation entropy is a lower bound on the cross-entropy. This aligns with our intuition—the model cannot possibly be highly accurate when the binarisation process is very uncertain.

To test for the effect of including cross-interactions, we fit the fully-connected model and null models using 10-fold cross-validation. For each survey participant in the validation (holdout) split we calculate the mean difference in relative entropy between the fully-connected and null models. Figure 5.5 shows the mean relative entropy across individuals and spins on the null and fully-connected models (left) and the mean *difference* in relative entropy, averaged across survey participants (right).

We find, in the right-hand panel, that the fully-connected model leads to significant reduction in relative entropy averaged over survey participants ($p < 0.05$). While statistically significant, the reduction is small. Since the measurement is averaged across survey participants, this finding suggests that the self-interaction model is sufficient to explain the behaviour for most participants—reflecting the slow-moving dynamics of the observed belief system—but not all participants. We anticipate that individuals whose belief states change more between observations are better explained by the fully-connected model than the null model.

Figure 5.6 explores this hypothesis, displaying the empirical probability density and cumulative distribution functions for the mean difference in relative entropy, across individuals. Indeed, we see substantial variation in effects. The weak right tail and heavy left tail indicate cases which are better-explained by the null and fully-connected models respectively.

We examine a sample of cases from either tail (top panel: left tail, bottom panel: right tail) in Figure 5.7. Each panel displays the change in binarisation probability between the two observed survey waves for a sampled survey participant. In the first three panels of the top row, we observe scenarios in which most spins are initially aligned, and a subset of the remaining spins then update to align with this set, i.e., where the system shifts toward a more consistent state. Conversely, in the first, second, and fourth panels

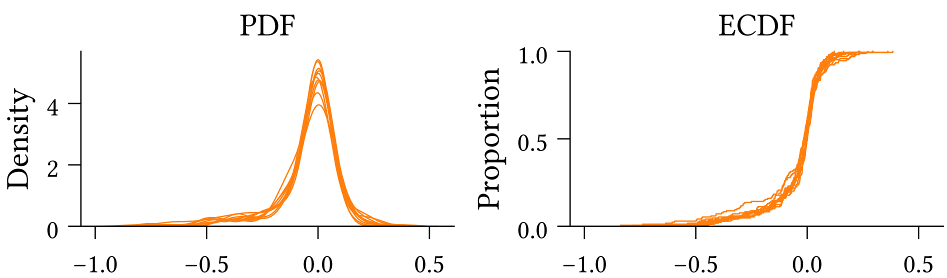


Figure 5.6: Probability density function (left) and empirical cumulative distribution function (right) of the mean difference in relative entropy between the fully-connected and null (only self-interaction) models across survey participants. Negative values indicate lower relative entropy for the fully-connected model.

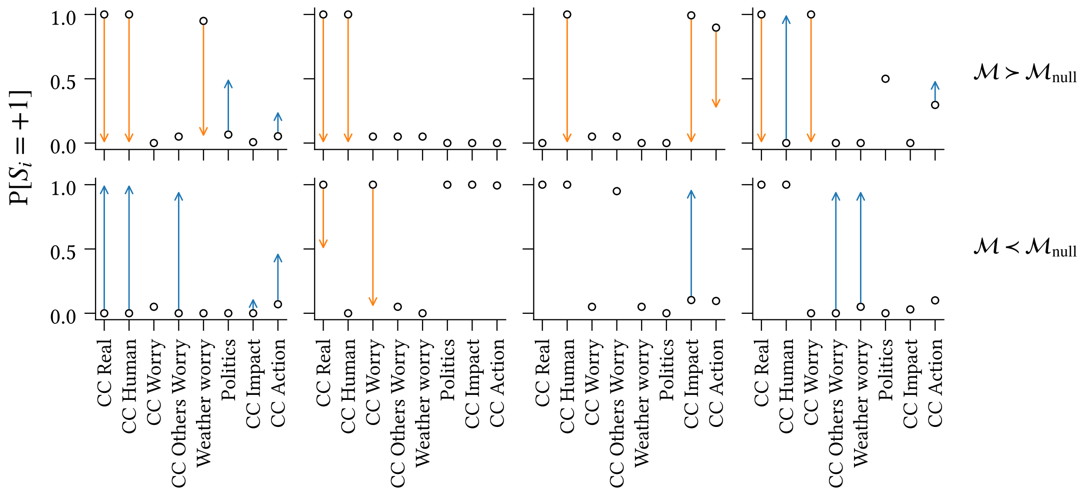


Figure 5.7: Observed transitions for sample survey participants in the left tail (top row) and right tail (bottom row) of Figure 5.6. Circles indicate the probability that participants' observations in the first wave are binarised to +1. Arrows denote the change in the second wave, colour-coded according to whether the change is toward the positive (blue) or negative (orange) state.

in the bottom row we see the opposite scenario play out. That is, the system is initially well-aligned, yet some spins update to a less consistent state.

The observed behaviour aligns with our expectations, namely that the model with cross-interactions outperforms the null model in scenarios where the belief system state updates toward a more consistent state (according to the theory of cognitive dissonance), and is out-performed when participants' behaviour contradicts this theory.

Chapter 6

Existence and impact of asymmetry in belief systems

To-do: Observations for outbound/inbound effects in Chapter 6.2.

Belief system models based on the Ising model, inspired by the cognitive dissonance theory of belief system dynamics, demonstrate impressive descriptive and explanatory capacities for various observed cognitive phenomena at both individual and collective scales. The currently prevalent view within the modelling community is that influence relations between beliefs or attitudes are symmetric — the pressure for X to align with Y is equal to that placed on Y to align with X. The present chapter places this assumption on trial.

We begin, in Chapter 6.1, by calibrating the non-equilibrium belief system model to the climate beliefs dataset described in Chapter 9.3. We demonstrate that belief system relations cannot be universally classified as symmetric, that some influence relations are recognisably asymmetric, and that relational asymmetry can vary in degree. In Chapter 6.2, we then show that this finding is not inconsequential. Asymmetric models exhibit different dynamics in experiments on belief-level interventions, with the potential to affect conclusions drawn about the relative effectiveness of interventions.

6.1 Existence of asymmetric relations

To investigate the existence of asymmetric relations in the asymmetric model calibrated to the climate beliefs dataset, we examine the differences in directional interaction effects for the asymmetric models calibrated using bootstrapping in the previous section.

For each bootstrapped model, $\mathcal{M}_{(i)}$ with parameters $\langle \boldsymbol{A}, \boldsymbol{J}_{(i)}, \boldsymbol{h}_{(i)} \rangle$, we obtain an estimate for the directional differential matrix:

$$\Delta_J^{(i)} = \boldsymbol{J}_{(i)} - \boldsymbol{J}_{(i)}^T \tag{6.1}$$

Recall that the k'th row of $\boldsymbol{J}_{(i)}$ (for $k \in \{1, ..., N\}$) describes the strength and direction of influence *from* the spin S_k *toward* each other spin. Hence for $k, \ell \in \{1, ..., N\}$ we should interpret the element $\Delta_J^{(i)}|_{k,\ell}$ of the directional differential matrix as the excess influence of spin S_k on S_ℓ. A positive value indicates that S_k exerts greater influence on S_ℓ than S_ℓ does on S_k.

Figure 6.1 shows the median directional differential for each pair of spins in the asymmetric model, in decreasing order. The 90% confidence intervals are calculated as the 5th and 95th percentiles across bootstrap samples of the directional differential matrix elements. For each pair of spins we display only the directional differential for

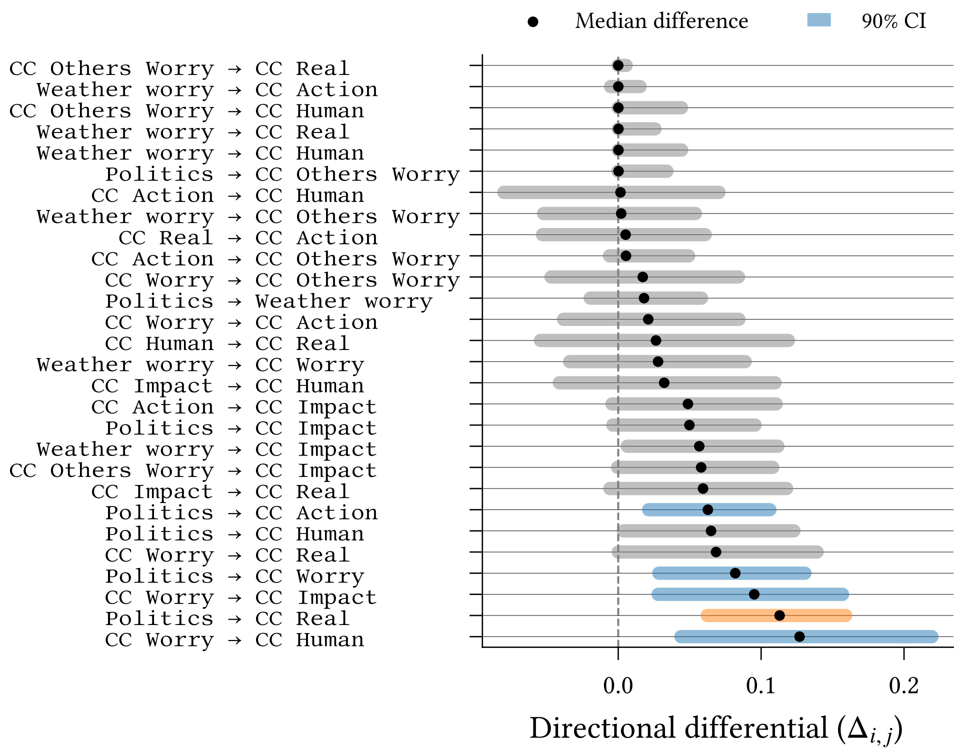


Figure 6.1: Directional differentials—the difference between directed interaction effects for a pair of spins—in reverse order of median value. Confidence intervals are calculated using the percentile method on models calibrated using bootstrapping ($n = 500$). Grey confidence intervals indicate absence of significant asymmetry. For cases with significant asymmetry, the colour indicates that one (**orange**) or both (**blue**) of the directed interacton effects are nonzero.

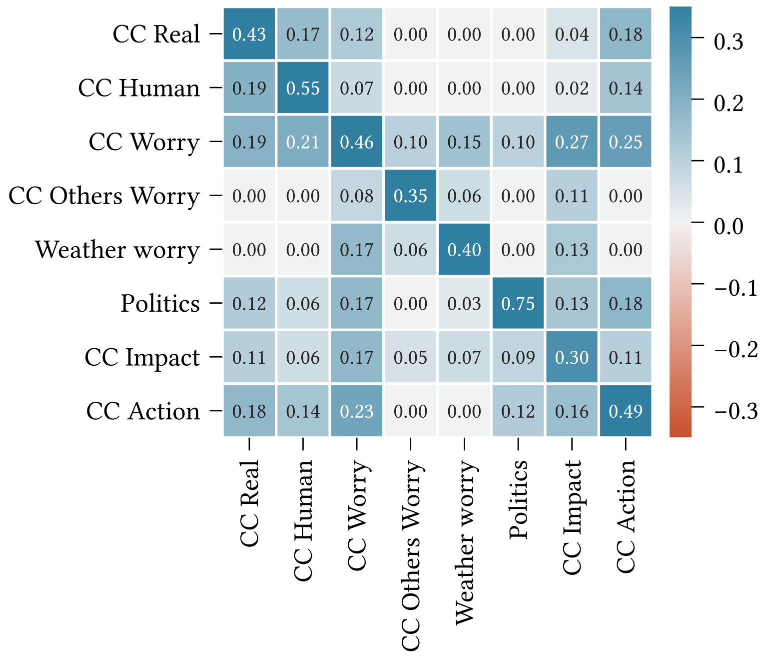


Figure 6.2: Interaction effect matrix ($\boldsymbol{J}$) for the asymmetric belief system model calibrated to the full climate beliefs dataset. See Figure 5.1 for original figure.

which the median is positive (since Δ_J is symmetric), and we exclude diagonal entries (which are zero by definition).

In Chapter 5 we cautioned against using bootstrapped confidence intervals to test for the *existence* of edges by comparison with zero when regularisation is used during calibration—this caution does not apply to the *comparison* of edge weights via the mean difference (Epskamp, Borsboom, et al., 2018).

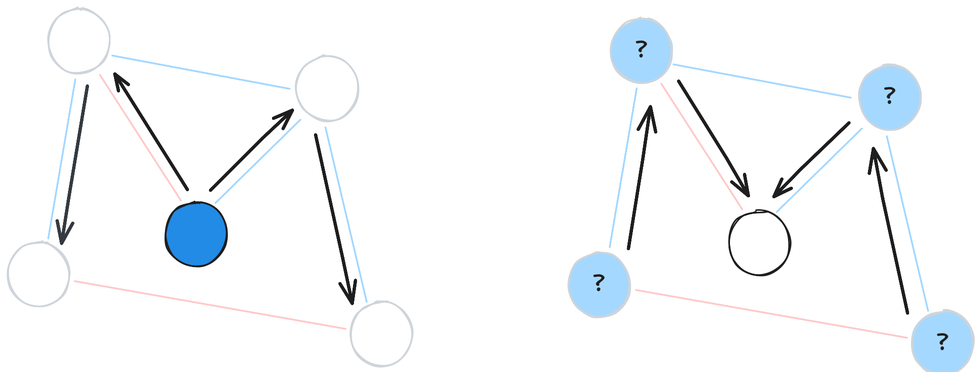
Figure 6.1 shows a majority of symmetric relations between spins (characterised by confidence intervals containing zero), with a small number of asymmetric relations. Most directional differentials display substantial uncertainty, with large confidence intervals (> 0.1 on average), reflecting the uncertainty in parameter estimates seen in the previous chapter (Figure 5.2).

We can partition the asymmetric relations into two groups through examination of the interaction matrix for the model calibrated on the full dataset (Figure 6.2, also shown in Figure 5.1 of the previous chapter). The first group comprises pairs of spins for which both interaction effects are nonzero, shown with blue confidence intervals: $\texttt{Politics} \to \{\texttt{CC Action}, \texttt{CC Worry}\}$ and $\texttt{CC Worry} \to \{\texttt{CC Impact}, \texttt{CC Human}\}$. The second group comprises pairs for which the interaction effect is nonzero in only one direction, shown with orange confidence intervals: $\texttt{Politics} \to \texttt{CC Real}$.

6.2 Asymmetry affects intervention dynamics

We now investigate differences in belief system behaviour under intervention for symmetric and asymmetric models calibrated to the climate beliefs dataset. In this section we are concerned with *collective* impacts, e.g., the proportion of individuals whose behaviour shifts due to an intervention. This reflects population-level intervention scenarios, such as media campaigns or policies which affect most individuals. We will return to *individual* impacts of intervention in the following chapter (Chapter 7.1).

We consider both **outbound** and **inbound** interventions, illustrated below. Outbound intervention experiments (left) examine how interventions on a particular spin propagate to other beliefs and attitudes. On the other hand, inbound intervention experiments (right) consider a single belief or attitude as the intervention target, and examine the differences in effects on this spin when intervening elsewhere in the network. We refer to the belief or attitude on which an intervention is applied as the **point-of-intervention** and the belief or attitude whose resulting state is measured as the **target**.



For outbound intervention experiments we examine the effects of intervening on the following spins:

- `Weather worry`: Level of worry about future extreme weather events
- `CC Worry`: Level of worry about current and future climate change
- `Politics`: General political party identification and ideology

These choices are motivated by our findings in the previous section. Recall that `CC Worry` and `Politics` both feature in the set of asymmetric relations, including in the relation between these two variables. `Weather worry` does not appear in any significant asymmetric relations, and—in comparison with the other variables—has relatively small outgoing interaction effects.

For inbound intervention experiments we consider interventions seeking to influence the `CC Action` variable, which captures individuals' general support for pro-environmental action on climate change (e.g., specific policies, support for increased regulation, and affirmative attitudes toward individual responsibility on climate change).

We define two quantities of interest: the **effect of intervention**, and **effect of asymmetry**. These are used, respectively, to assess an intervention's impact on a given spin's behaviour with reference to our expectations in a no-intervention scenario (Definition 6.2.1), and the difference in a spin's behaviour in the asymmetric model, compared to our expectations in the symmetric model (Definition 6.2.2). Note that the effect of asymmetry is not inherently concerned with intervention, but is a general measure for the difference between asymmetric and symmetry model dynamics.

Both quantities compute a difference in effects between two distinct models: the intervention and null models in the case of the effect of intervention; the asymmetric and symmetric models in the case of the effect of asymmetry. To ensure comparability of results, difference are always computed between models with identical random number generation contexts.

Definition 6.2.1 (Effect of Intervention) Let $f_{\mathcal{M}}(\boldsymbol{s}_0; t)$ be the stochastic function which returns the state $\boldsymbol{s}^t$ resulting from simulation of the model $\mathcal{M}$ with initial state $\boldsymbol{s}_0$.

For an intervention model $\mathcal{M}_\delta$ with arbitrary point-of-intervention, we define the **effect of intervention** for $\mathcal{M}_\delta$ as the change in outcome with respect to the null (no-intervention) model, $\mathcal{M}_0$:

$$f_{\mathcal{M}_\delta}(\boldsymbol{s}_0; t) - f_{\mathcal{M}_0}(\boldsymbol{s}_0; t) \tag{6.2}$$

Definition 6.2.2 (Effect of Asymmetry) Let $f_{\mathcal{M}}(\boldsymbol{s}_0; t)$ be defined as in Definition 6.2.1. We define the **effect of asymmetry** for an asymmetric model, $\mathcal{M}^{\text{asym}}$, as the observed change in outcome with respect to the corresponding symmetric model, $\mathcal{M}^{\text{sym}}$:

$$f_{\mathcal{M}^{\text{asym}}}(\boldsymbol{s}_0; t) - f_{\mathcal{M}^{\text{sym}}}(\boldsymbol{s}_0; t) \tag{6.3}$$

In each of the intervention experiments described below, we use the models calibrated to the full climate beliefs dataset (Chapter 5). For a given experiment, we initialise the associated model (symmetric/asymmetric; intervention/null) with the binarised observations from the final wave of the climate beliefs dataset. We then draw consecutive samples from the model using parallel glauber dynamics (Chapter 3.4) until $t = 5$, before measuring the state of the target spin. Since the duration between waves in the climate beliefs dataset is approximately six months, the choice of $t = 5$ corresponds to measuring the impact of intervention after roughly two and a half years in the model timescale. To account for stochasticity in both the binarisation and simulation procedures we perform 500 repeats of each intervention.

We first consider the impact of intervention strength on effect of intervention. The strength of an intervention affects the degree to which the intervention changes the state of the point-of-intervention. However, this is also sensitive to the existing influence on that spin[6]. Since the climate beliefs dataset comprises individuals with different states, it follows that intervention strength may impact not only the degree to which the behaviour of a target spin changes, but also the relative effects of different interventions.

To understand the potential impacts of intervention strength, we compare the mean effect of intervention across individuals for intervention strengths $\delta_h \in \{0.5, 2.5\}$, for each of the outbound points of intervention listed above Figure 6.4. The choice of intervention strengths is motivated by our earlier analysis in Chapter 3.5, with the two strengths corresponding to 'weak' and 'strong' interventions respectively, as judged by their direct impact on the behaviour of the point-of-intervention.

We observe a strong linear relationship between intervention strength effects for each variable, indicating that—at the population-mean level—the impact of intervention

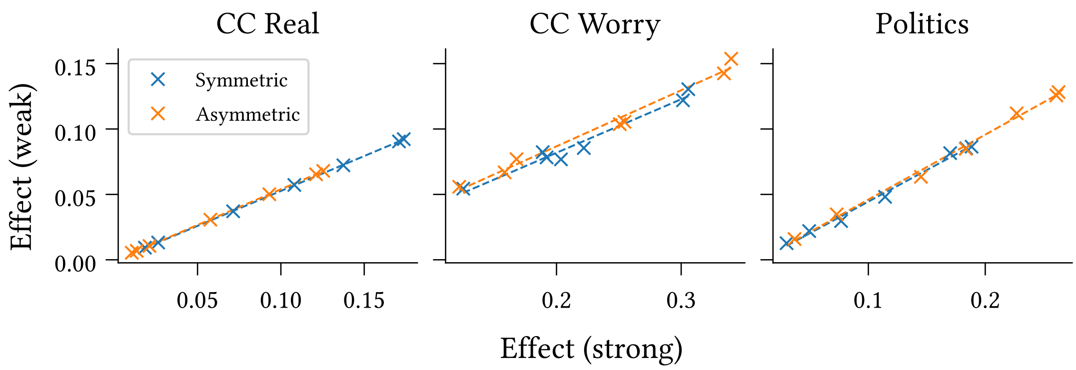


Figure 6.4: Weak ($\delta_h = 0.5$) and strong ($\delta_h = 2.5$) interventions exhibit strong positively correlated effects of intervention, consistently across points-of-intervention (panels) and intervention targets (scatterplots). This finding holds for both symmetric and asymmetric models calibrated to the climate beliefs dataset.

[6]For detailed discussion on the selection and implications of intervention strengths see Chapter 3.5.

strength on effect is mostly one of scale, that applies similarly to all target spins. For the purposes of the following experiments it then suffices to consider only a single intervention strength. As our current focus is on intervention propagation and the effects felt at spins other than the point-of-intervention, we take $\delta_h = 2.5$, such that the interventions can be considered generally effective at shifting the behaviour at the point-of-intervention.

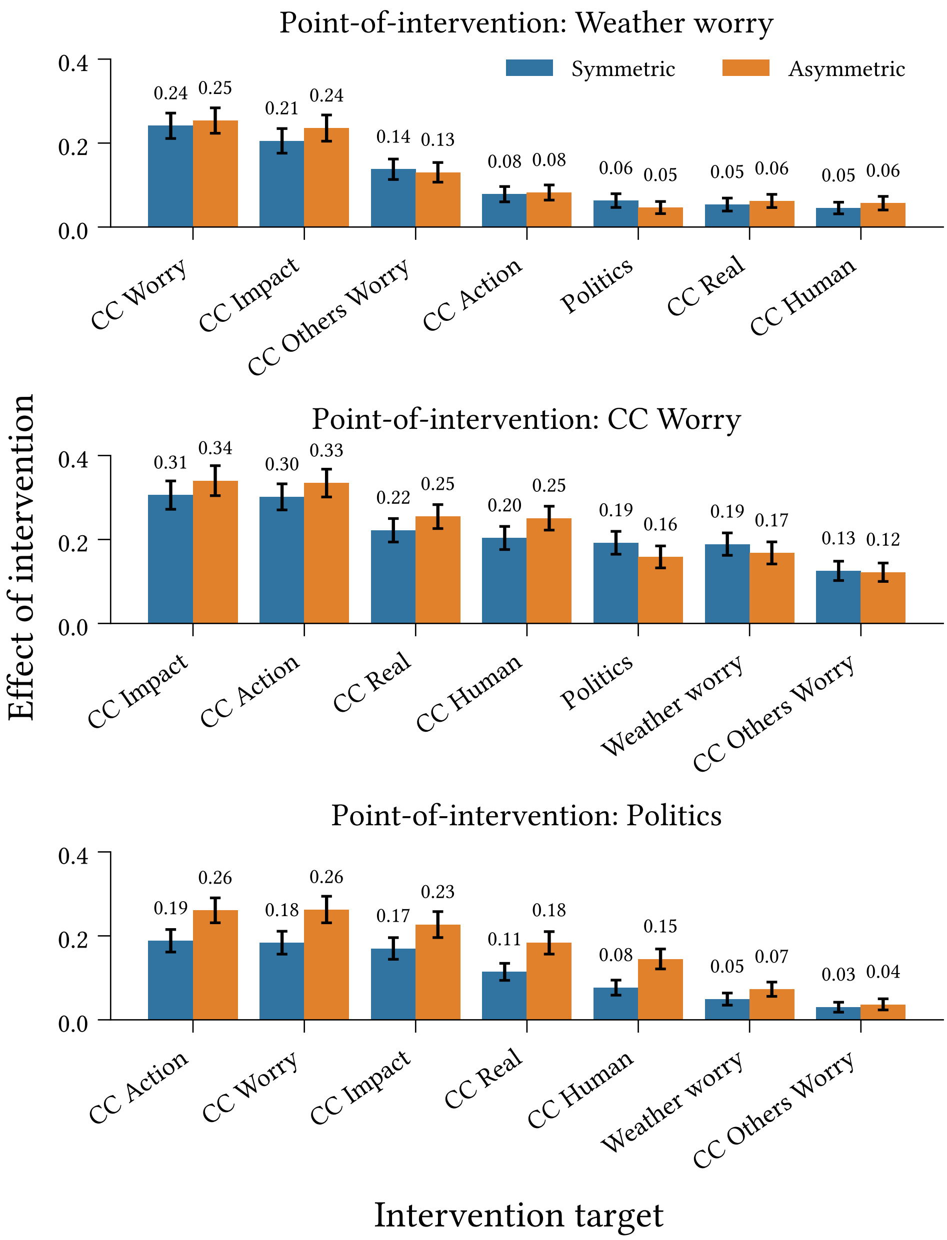


Figure 6.5: Outbound effect of intervention (Definition 6.2.1) at each target spin for interventions targeting `Weather Worry`, `CC Worry`, and `Politics`, in symmetric and asymmetric belief system models calibrated to the climate beliefs dataset. Intervention strength:w $\delta_h = 2.5$. Confidence intervals display 1.96 standard deviations around the mean effect, measured across 500 repeated simulations.

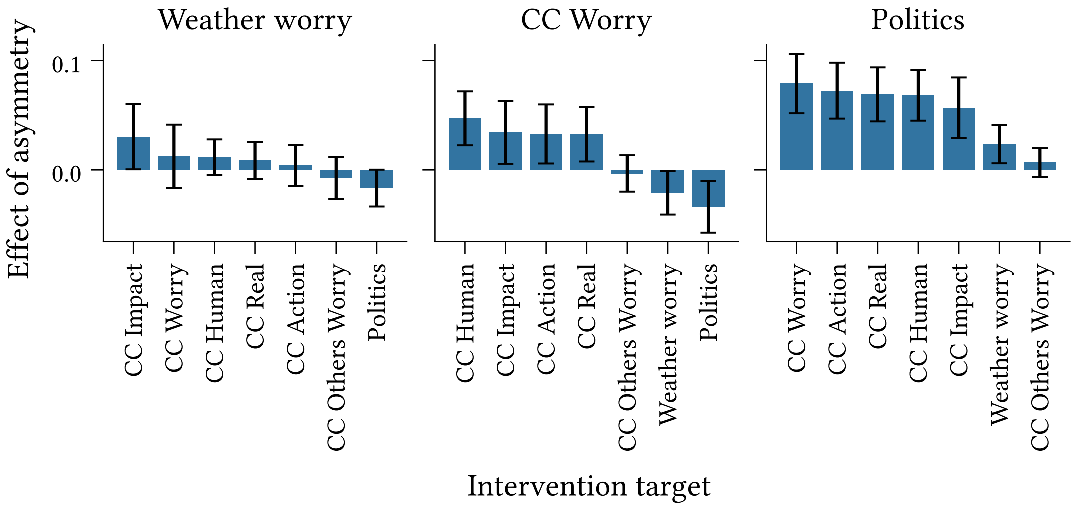


Figure 6.6: Outbound effect of asymmetry (Definition 6.2.2) at each target spin for interventions on `Weather Worry`, `CC Worry`, and `Politics`, for models calibrated to the climate beliefs dataset. Intervention strength: $\delta_h = 2.5$. Confidence intervals display 1.96 standard deviations around the mean effect of asymmetry ($n = 500$).

Figures 6.5 and 6.6 show the outbound mean effects of intervention and asymmetry, respectively, across individuals, for the points of intervention listed above. The confidence intervals show two standard deviations around the mean values.

We observe, in Figure 6.5, that different targets exhibit different effects of intervention. The effect generally decreases with the magnitude of the interaction from the point-of-intervention to the target spin; however, we also note that all effects are positive. This suggests that for the measurement time examined here, interventions act primarily through direct interactions with the target spin, with some indirect influence propagating through intermediary interactions. For instance, while `Weather Worry` has no direct influence on `CC Action`, we see nonzero effect of intervention as a result of indirect paths, e.g., through `CC Worry`. We see further evidence of this through comparison with an analogous figure (see Figure B.1 in Appendix) using a measurement time of $t = 10$ (approximate five years in the models' timescales), which shows larger increases in the effect of intervention for variables with stronger incoming interactions.

While the confidence intervals for the symmetric and asymmetric models overlap significantly for most targets in Figure 6.5 (with some exception in the bottom panel), the effects of asymmetry displayed in Figure 6.6 provide a cleaner comparison. We observe no significant differences between the two models for the `Weather Worry` scenario. For the remaining two scenarios, however, we see several significant differences. In particular, all pairs of variables with asymmetric direct relations—identified in the previous section—display significant differences.

Notice that while `CC Worry` exhibits greater intervention effects than `Politics` in Figure 6.5, `Politics` exhibits larger mean effects of asymmetry in Figure 6.6 (though the latter differences are not statistically significant). Comparison of the interaction effect matrices (Figure 5.1) shows direct connectivity differences for `Politics` between the models. While the symmetric model excludes interactions with `CC Human` and `Weather Worry` entirely, the asymmetric model permits *outbound* interactions with these spins, allowing interventions to propagate. By contrast, interventions on `Politics` in the symmetric model are more reliant on indirect propagation.

Turning our attention toward inbound intervention dynamics, we now consider the effects of different interventions seeking to influence the state of `CC Action`. The top panel of Figure 6.7 shows the mean effect of intervention observed at the target spin for each possible point-of-intervention. Examining the mean effects allows us to gauge the expected absolute effect of each intervention, but is limited for purposes of comparing the relative effects of interventions.

The mean effect of intervention, calculated separately for each point-of-intervention, erases information regarding the expected relative effectiveness of different interventions.

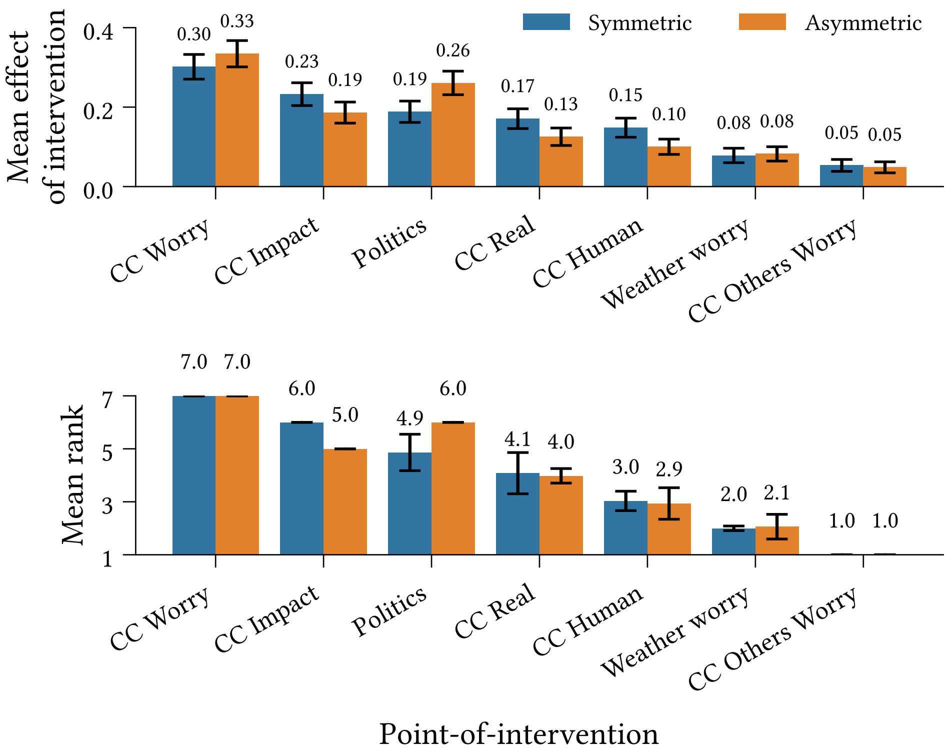


Figure 6.7: Inbound mean effect of intervention (Definition 6.2.1) (*Top*) and mean rank (*Bottom*) for interventions targeting `CC Action`. Ranks are calculated per-repeat with respect to the average effect of intervention—higher values are assigned higher ranks. Confidence intervals display 1.96 standard deviations ($n = 500$).

To assess relative effectiveness, we therefore also estimate the expected ranking over points-of-intervention (the bottom panel of Figure 6.7). For each repeat, we order the points-of-intervention in increasing order of average collective effect of intervention. Higher values are assigned higher ranks. In the case of a tie, we assign all tied spins the minimum rank which would have been assigned to the group, had they been distinct[7].

As with the outbound case, we find that the effect of intervention on `CC Action` varies depending on where we intervene. Points-of-intervention (along the x-axes) are sorted in decreasing order of mean values. We observe identical orderings between the effect of intervention and ranking plots, showing good correspondence between the two measures. We observe a clear inversion in this ordering for the asymmetric model, with `Politics` exhibiting the second-highest rank, and (at least) second-highest mean effect of intervention ($p < 0.05$).

Unlike with the outbound effects, the order of effect sizes does not align well with the corresponding strengths of direct interactions strengths into the target variable. For instance, `CC Real` and `CC Human` have the second- and third-strongest cross-interactions into `CC Action` in the symmetric model, yet the fourth- and fifth-highest effect of intervention. This likely reflects the low outbound connectivity of these variables in comparison with `CC Impact` and `Politics`, despite the latter spins having relatively lower direct on the target—additional evidence that indirect influence plays a critical role in intervention dynamics.

Most confidence intervals around the mean ranks are small, indicating negligible (if any) variation between simulations or possible binarisations of the dataset. However, we do observe three larger, overlapping confidence intervals for each model. These arise due to variation the mean effect of intervention causing occasional inversions of the ranking order. Note that we only observe inversions between points-of-intervention which are adjacent in the sorted order.

[7]For instance, suppose we have five variables, $\{A, B, C, D, E\}$ with average collective effects $\{0.2, 0.2, 0.6, 0.4, 0.1\}$. Then the assigned ranking would be $\text{Rank}(A, B, C, D, E) = [2, 2, 5, 4, 1]$. The variables B and C are both assigned the rank 2. No variable is assigned rank 3.

Chapter 7

Heterogeneous belief systems and intervention effects

In this chapter we investigate individual heterogeneity in both intervention behaviour and belief system structure in asymmetric, non-equilibrium belief system models.

While the previous chapter was primarily concerned with the *collective* effects of intervention, the distribution of expected effect of intervention across individuals for inbound interventions targeting 'Climate Action' shows substantial variation between survey participants (Figure 7.1). In Chapter 7.1 we investigate the conditions under which a given intervention targeting attitudes on climate action is likely to be effective.

Our subsequent investigation in Chapter 7.2 considers how inferred belief system structure varies between individuals, or subgroups in a population. In the previous

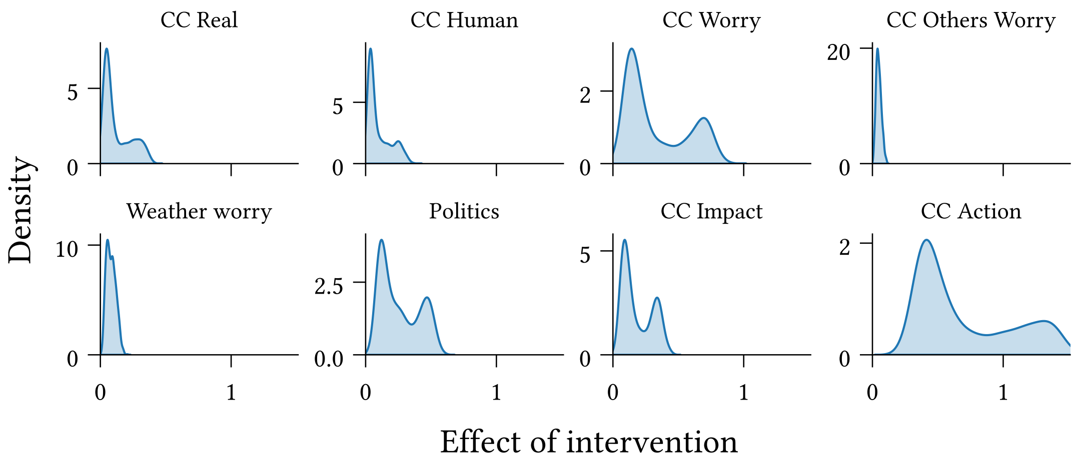


Figure 7.1: The expected effect of intervention for strong interventions ($\delta_h = 2.5$) targeting 'Climate Action' exhibits substantial variation between individuals.

chapter we identified political ideology as both exhibiting significant asymmetric influence on other variables (Chapter 6.1) and an effect point-of-intervention for interventions looking to affect attitudes on climate action (Chapter 6.2). Here, we investigate whether this influence extends beyond pairwise interactions with other variables by comparing asymmetric belief systems calibrated to the (self-reported) liberal and conservative subpopulations within the climate beliefs dataset.

7.1 Heterogeneity in intervention effects

In this section we investigate the conditions under which interventions targeting the 'Climate Action' attitude are likely to be successful, in asymmetric non-equilibrium models calibrated to the climate beliefs dataset.

The intervention simulation procedure is identical to that described in the previous chapter. For each survey participant and point-of-intervention, we collect 500 repeated samples of the 'Climate Action' spin state at $t = 5$. We simulate both *weak* and *strong* intervention scenarios, $\delta_h \in \{0.5, 2.5\}$.

We first examine variation in the rankings over points of intervention between different individuals. For each participant, we estimate the expected effect of intervention by taking the average across repeated samples for each point-of-intervention and scenario. We then rank the points of intervention by expected effect, such that for each participant and scenario, we obtain a ranking over possible points of intervention. Figure 7.2 shows the average rank across individuals for each possible point-of-intervention and scenario. The whiskers denote one standard deviation around the average.

TODO: Observations

We now proceed to characterise the conditions under which each intervention is effective. Recall that the model's initial state is the only distinguishing factor between survey participants in the intervention simulations. It follows, therefore, that any difference in intervention effectiveness between participants results from differences in their initial states. To characterise the conditions for successful intervention, it then suffices to characterise the set of *initial states* which lead to effective interventions, and distinguish them from those that do not.

We fit a shallow decision tree regression model for each point-of-intervention, taking the expected intervention effect for each survey participant as the response variable, and the non-binarised measurements from the final wave of the climate beliefs dataset as features. The parameterised model partitions the space of possible initial states,

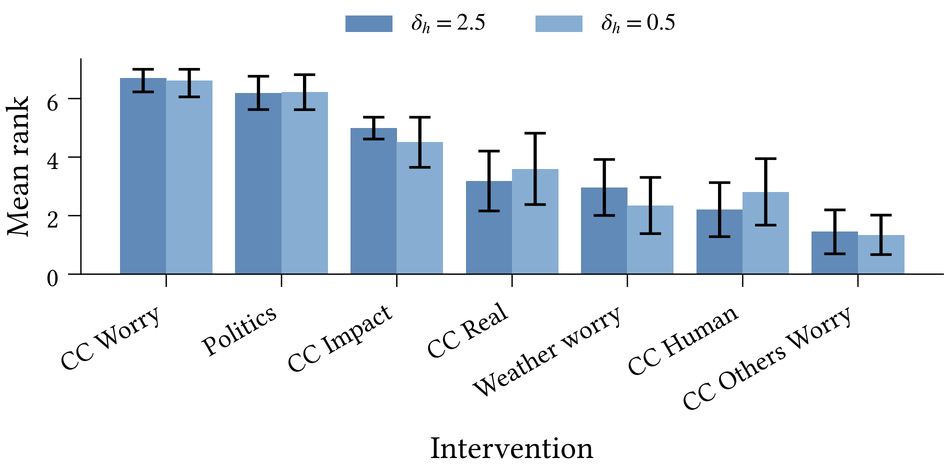


Figure 7.2: Variability in point-of-intervention ranks across climate attitudes survey participants, for interventions targeting 'Climate Action'. Bar height describes mean rank across individuals; whiskers denote one standard deviation around the mean. Bar colours indicate strong ($\delta_h = 2.5$) and weak ($\delta_h = 0.5$) intervention scenarios.

assigning a prediction value to each region, such that the mean-squared error with respect to the observed intervention effects is minimal. We characterise the *personas* which lead to effective interventions using the rules which define high-effect regions of the initial state space.

For this experiment, we use a tree depth of 3 so as to generate short rules[8], and classify intervention effects within the upper quartile of observed values as 'high effect'. We only consider personas which are sufficiently prevalent in the observed data ($\geq 15\%$).

Figure 7.3 shows the high-effect personas for each intervention scenario and point-of-intervention, excluding points of intervention for which there are no sufficiently high-effect observations, where the upper quartile is below 0.1. Each row describes the results for a distinct point-of-intervention, indicated at the right of the figure. Let us step through the remaining columns:

- The left-most column displays intervention effect distribution for the strong intervention scenario, with upper quartile regions indicated.
- The central column describes the set of personas predicted to be most susceptible to the given intervention. Each row describes a distinct persona. Each cell described a subinterval of the range of possible feature values, $I \subseteq [-1, 1]$, with 'L' referring to 'Low' and 'H' to 'High'. The modifiers 'V' and 'S' (not present) refer to 'Very' and

[8] A tree with depth $n \in \mathbb{N}$ partitions the initial state space into 2^n regions, each described using n feature rules.

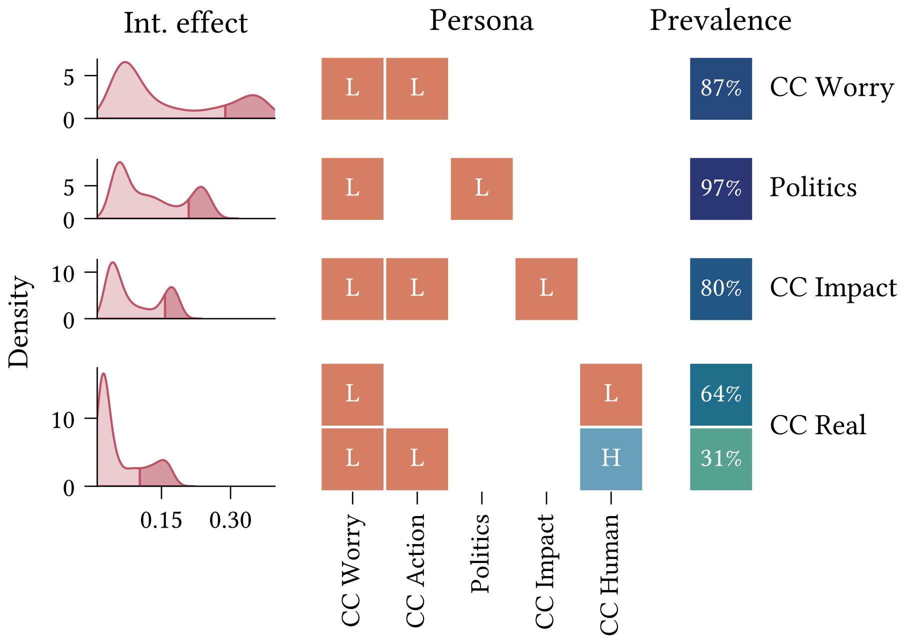


Figure 7.3: **TODO**

'Slightly', respectively. Entries with the for ' X' refer to the complement of the interval named by X. The specific interval mappings are defined in **REFERENCE TABLE**.

- The rightmost column describes the prevalence of each persona within the 'high-effect' population. Cell values are the proportion of individuals within the upper quartile for each scenario who satisfy the persona. Missing entries indicate that a persona was either not identified for the given scenario, or had prevalence below 15%.

TODO: Observations

7.2 Heterogeneity in belief system structure

- **Overall goal of these experiments:** Understand how the belief system inferred from the full climate beliefs dataset may differ from those inferred from subsets.

- **In broad strokes, how do we test this?**
  - We partition the climate beliefs dataset into self-reported conservative and liberal individuals. We then calibrate separate asymmetric belief system models for each group.
  - We compare the observed belief system structures, and contrast with the belief system calibrated to the full dataset.

- **Specific experimental details:**

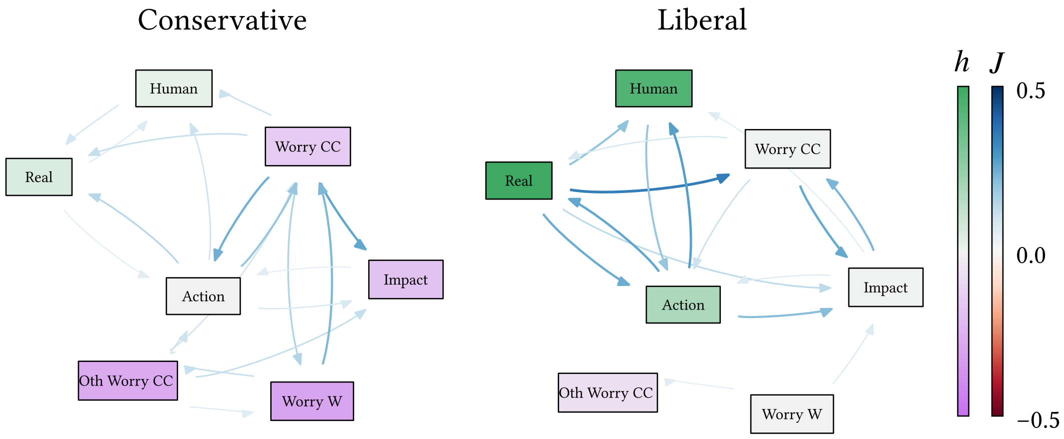


Figure 7.4: Prefixes: A (attitude), B (belief); node labels: CC (climate change), CCA (climate change anthropogenic), CCW (climate change worry), CCWO (climate change worry others), CCI (climate change impacts), CCP (climate change policies), WW (weather worry).

- Since we subset the data by political ideology, we remove the `Politics` variable from the climate beliefs dataset. So the models are calibrated with only seven spins, as opposed to eight.
- We apply regularisation to the model calibration, with regularisation strength determined independently for each group (Table 4.1).

- **Results & figures:**
  - Belief system networks

Chapter 8

Discussion

8.1 Plan

- Draw out the key findings from the above two results sections
- Make clear the implications for our theory of belief system dynamics
- Sensitivity of parameter estimation to unmeasured factors or incorrect structural assumptions. i.e., what happens when we cannot include an influential belief, or when we falsely assume that a relation does or does not exist?
- Representational limitations of our model:
  - Pairwise relations limit what can be modelled. Demonstrate using vaccination example — we can't capture relations between pairs of attitudes, whose effect sign or magnitude depends on a third belief/attitude.
  - Polar $(-1, +1)$ spin state assumption; some beliefs may be better treated as 'on/off', where the 'off' state has *no* effect on other spins, rather than a negative effect.
  - More generally, effect sizes may vary depending on specific spin states.

Chapter 9

Climate beliefs dataset

In this chapter we outline the context and construction of the dataset used for model calibration in Chapter 4. We will first describe the broader context and complexities of the dataset underlying this study, before outlining our data validation and cleaning methods in Chapter 9.2. Finally, in Chapter 9.3 we detail the construction of the targeted dataset used for model calibration (Chapter 4).

While many empirical studies on belief systems rely on cross-sectional data (Lee et al., 2025; Powell et al., 2023; van Noord et al., 2025), longitudinal data is necessary to capture changes in individuals' internal cognitive states over time. In fact, repeated per-individual observations are essential to our adopted method for model calibration. We are fortunate, in this study, to have had access to a longitudinal dataset on climate beliefs and attitudes in USA as our primary data resource.

The longitudinal study (**CITE**) comprises six waves of responses from individuals residing in the United States, collected between April 2020 and March 2023. The assessed dimensions include general demographic information (e.g., age, gender, education, and financial status), beliefs, attitudes, and experiences relating to concurrently-salient topics such as COVID-19, climate change, or the 2020 US presidential election, and support for hypothetical policies. In addition to a wide-form table of survey response data, the dataset includes a codebook which specifies, for each survey item, the question text, occurrence in survey waves, and conditional display logic where applicable. At present, the codebook is limited to Waves 1—5, i.e., excluding the sixth (final) wave.

However, the dataset is not without complexities. Survey participation varies, with each participant responding to a (possibly non-contiguous) subset of waves. Figure 9.1 shows the number of participants who have responded to each combination of survey waves, with a minimum count of 1500. While each individual wave has a relatively

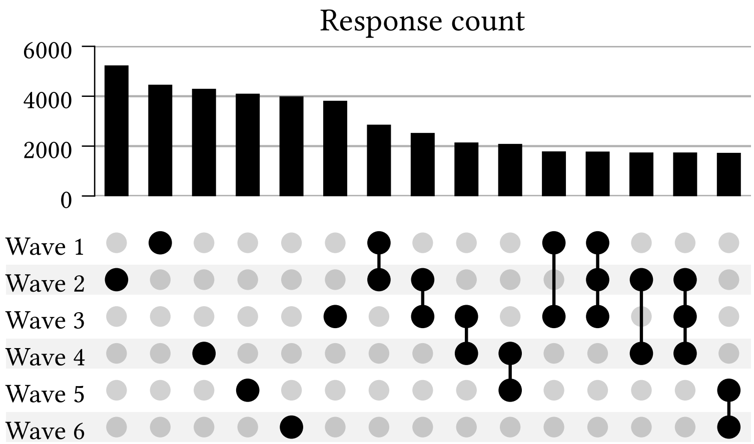


Figure 9.1: Number of repeat participants for different wave combinations. Combinations with total responses below 1500 have been removed.

high response rate (roughly 4000–5000), this rapidly drops off when other waves are considered. For instance, only ~2500 individuals responded to Waves 1 and 2, and only ~1900 of these individuals also responded to Wave 3.

Survey content also varies between waves and participants, both with regards to *which* questions are included, and *how* they are presented. Certain questions are displayed only in particular waves, or only to either repeating or new participants. Some questions are shown conditionally based on an individual's responses to prior questions (within the same wave). Others vary according to survey treatment conditions, such that individuals in different treatment groups are presented different variants of a question. This survey logic is not always executed correctly; in some cases participants are shown survey questions incorrectly (e.g., 'new' participants shown questions intended only for 'repeating' participants). We discuss this issue further in Chapter 9.1.

Question format, text, and response schemas also occasionally change between survey waves. Changes in response schema are less common, however, and typically affect items with categorical responses. With a view to constructing the calibration dataset for our experiments, we note that most response schemas are not binary and therefore require binarisation (see Chapter 4.2).

Casting our attention to the survey timing, we examine the survey response dates for each wave (Figure 9.2) and the distribution of interval durations between consecutive-wave responses across individuals (Figure 9.3), i.e., the 'inter-response time'.

In Figure 9.2 we observe that the survey waves occur with irregular spacing and duration. Some consecutive pairs of waves are much closer than others. For instance, notice that:

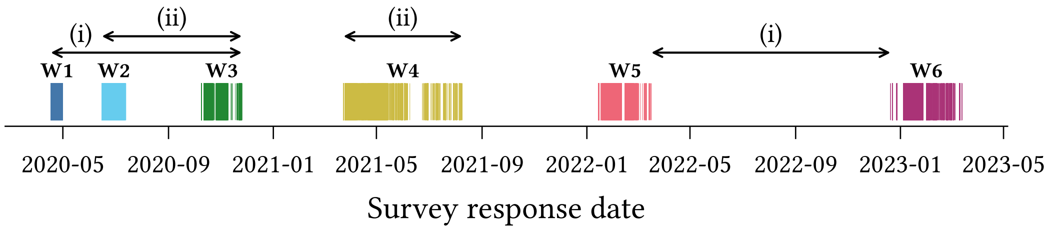


Figure 9.2: **TODO**

(i) The duration from the start of Wave 1 until the end of Wave 3 is shorter than the duration *between* Waves 5 and 6, and

(ii) The duration spanned by Wave 4 alone exceeds that of the interval spanned by both Waves 1 and 2.

This poses a potential problem for model calibration, since the non-equilibrium belief model (defined in Chapter 3) operates on the assumption that samples are equispaced.

We see the irregular spacing reflected also in the inter-response time distributions. For each pair of consecutive waves, we observe substantial variation in the time between responses for different participants.

Finally, we note that the time interval spanned by the longitudinal dataset includes several notable events which could reasonably be expected to influence — and confound — the dynamics of beliefs and attitudes in myriad contexts. These include the COVID-19 pandemic, which arrived in the US only three months prior to the first survey wave (Holshue et al., 2020), the 2020 US Presidential Election which occurred during Wave 3, the January 6 United States Capitol Attack, which occurred between Waves 3 and 4, ...

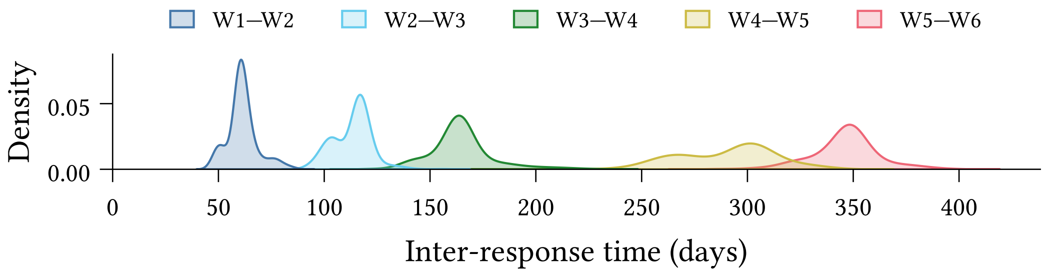


Figure 9.3: Between-response time distribution

9.1 Data Validation

The complexity of the climate attitudes survey, as outlined above, necessitates a rigorous approach to data validation, both to identify errors in the expected schema and to ensure that the data matches our expectations.

We have implemented a general validation pipeline comprising three stages:

1. **Type-level validation:** The set of columns is exactly as expected, and all columns have the correct data type.
2. **Response-value validation:** All *non-null* responses are valid according to the schema specified in the codebook.
3. **Null-value validation:** Responses are null if, and only if, we expect them to be null.

9.1.1 Type-level validation

Type-level validation ensures that the *observed* data schema matches the *prescribed* schema. Since response data types vary between questions (see Table 9.1), the type-checking must be flexible and capable of handling complex data types.

For instance, categorical multiple-choice responses are represented using a `list[Enum]`, where `Enum` is a question-specific enum-type[9]. Type validation for a multiple-choice question thus requires checking: (i) that the response column comprises `list`'s, and (ii) that all list elements belong to the set of values defined by the `Enum`.

Response schema	Raw type	Coerced type
Text	`str`	`str`
Single response (ordinal)	`int`	`int`
Single response (categorical)	`int`	`Enum`
Multiple response	`str`	`list[Enum]`
Numeric	`float`	`float | int`

Table 9.1: **TODO**

[9]An enum is a type defined by a finite set of allowable values. In our case the values are human-readable strings. For instance, the `dem_urban` survey question, which asks 'What kind of area do you live in?' has responses with data type described by the enum $\{\text{Urban}, \text{Suburban}, \text{Rural}\}$.

9.1.2 Response-value validation

Response-value validation then ensures that all non-null response values are valid according to the survey codebook. For most variables this is straightforward. Numeric and single-response ordinal variables typically have a defined range (e.g., $18 \leq \text{age} \leq 99$, or $1 \leq x \leq 5$ for a 5-point Likert scale variable). Text-entry responses are always considered valid.

Single-response categorical questions have `Enum` type, so type-level validation is sufficient to ensure that the responses do not contain any values outside the set defined by the `Enum`. However, the allowable values occasionally change between survey waves. Hence response-value validation is also required for categorical single-response and multiple-response variables, and in general must handle schema variation between waves.

9.1.3 Null-value validation

Finally, null-value validation ensures that responses are null if, and only if, they are expected to be null. For a question Q, the response of a participant P in wave W is permitted to be null, iff, at least one of the following four conditions is true:

N1. Q is not asked in wave W,

N2. In wave W, Q is only shown to *new* participants, but P is a *repeating* participant (or vice versa),

N3. In wave W, Q is only shown to a treatment group of which P is not a member, or

N4. In wave W, Q is conditionally shown to participants based on their response to a distinct question Q', and P's response to Q' does not satisfy the condition.

Formally we can write the condition for P's response R being null as:

$$\text{null}(R) \iff (\text{N1} \vee \text{N2} \vee \text{N3} \vee \text{N4}) \tag{9.1}$$

Null-value validation then consists in identifying the set of responses $\boldsymbol{R}$ such that for $R \in \boldsymbol{R}$, exactly one of $\text{null}(R)$ and $(\text{N1} \vee \text{N2} \vee \text{N3} \vee \text{N4})$ is true. Validation succeeds if $\boldsymbol{R}$ is empty. Otherwise the members of $\boldsymbol{R}$ are counterexamples to the above conditions.

9.1.4 Implementation

We implement type-level and simple response-value validation using the Pandera Python library for DataFrame validation (Bantilan et al., 2022). We implement manual validation checks for more complex response-value cases, such as questions whose response schema varies between waves.

The null-value validation logic, i.e., the process of testing for contradictions Equation 9.1, is also manually implemented. Conditions N1 and N2 are derived automatically from the survey codebook. Conditions N3 and N4 must be manually specified, as the codebook currently does not have a standardised method for describing conditional display logic.

9.1.5 Validation results

At the time of writing, approximately 25% of the complete survey question set for Waves 1–5 has been validated, including all survey questions which were considered for the targeted calibration dataset (Chapter 9.3). Since Wave 6 is currently undocumented in the survey codebook we can reasonably validate neither the data schema, nor the presence of null-values. We have therefore not yet validated any of the data from Wave 6, and exclude this wave from the present study.

All type-level and response-value validation checks succeed, providing a strong guarantee that the data schema matches our expectations per the codebook. We do, however, encounter several problems during null-value validation.

In some cases, these were due to errors in the codebook itself. These errors are relatively straightforward to identify from the null-value validation results, since they often affect all individuals in a particular wave. For instance, if the codebook specified that a question is not presented in Wave 1, yet all responses are non-null, then this most likely indicates an error in the codebook. While some cases are more subtle, such as where a treatment class is misspecified, when arising due to a codebook error we still expect the contradictions to affect a well-defined subset of the population (in this case, the individuals in a particular treatment class).

In other cases we identify contradictions which are due to errors in the survey process, which caused certain survey questions to be displayed to some individuals in conditions which did not satisfy the specified survey logic. Some such cases were systematic, affecting all individuals until the survey process was updated; however, in other cases we have not been able to identify the source of the error.

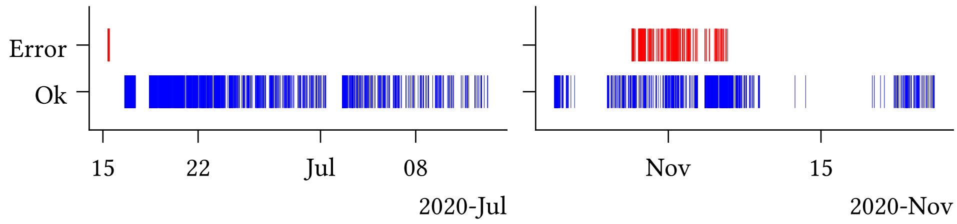


Figure 9.4: **TODO**

We exclude all responses which fail the null-value validation from further analysis in the data cleaning stage (Chapter 9.2.2).

The described validation process is primarily concerned with ensuring that the data schema — comprising types and values — aligns with our expectations as per the survey codebook. While comprehensive in this regard, the process does not account for *all* possible forms of errors or inconsistencies. One category which is currently unaccounted for, but is worth mentioning, comprises inconsistencies in the responses from a given individual over multiple waves. For instance, the dataset includes several cases in which individuals presented the question:

"Were you born in the United States?"

respond with 'Yes' in one wave, but 'No' in another. Validating data for inconsistencies such as this requires careful consideration on a question-by-question basis to assess response types for conflicts. We consider this category beyond the scope of this study, describing it here only to illustrate the bounds of the above validation process.

9.2 Normalisation, cleaning, and transformations

We apply pre-processing steps to make the dataset amenible to downstream analysis, which are divided into three categories:

- **Normalisation:** Ensuring the data schema (variable names, types, and values) conform to consistent standards.
- **Cleaning:** Filtering erroneous or otherwise-problematic data.
- **Transformation:** Structural dataset enrichments, including constructed data variables and improved organisation of information.

9.2.1 Normalisation

Variable names in the original dataset are subject to some inconsistency, featuring a mixture of `snake_case`, `CamelCase`, `dot.case`, and `kebab-case`. We therefore first convert the names of all variables specified in the codebook to a standard format, making all names lowercase, and replacing hyphens and periods with underscores.

The dataset also contains several columns which are not described in the codebook, often containing survey metadata, the functions of which are typically indicated using a variable name prefix. For instance, the `Group_` prefix indicates that a column describes treatment group membership. We do not normalise these variable names here, so as to maintain identiability for later transformations (Chapter 9.2.3).

Secondly, we coerce all variable data types to a standard set. In most cases this is straightforward. Dates and times represented as strings are coerced to timezone-free `datetime` types. Non-floating-point numeric values (which constitute most survey response types) are coerced to 64-bit integers.

Categorical variable responses are originally represented as integeres, which presents three issues. Firstly, it imposes an ordinal structure which does not, in general, make sense for categorical variables. For instance, consider the first four response options to a survey question asking participants about their recent experience with different forms of extreme weather:

1. Severe winter storm, 2. Hurricane, 3. Tornado, 4. Heat wave

One may devise any number of reasonable ordinal interpretations over this set (e.g., 'highest average risk of property damage', or 'annual emergency response cost'), yet such interpretations are context-dependent and not inherent to the categories themselves. Recording categorical responses using numeric types therefore risks spurious interpretations. Second, integer values provide no information about the categories to consumers of the dataset, requiring that they cross-reference manually with the codebook. Thirdly, in some cases the mapping from categories to integers for a given question varies between survey waves. We account for all three of these issues by coercing categorical variables to `Enum` types, which preserve information about the underlying categories, and do not assume an ordinal structure.

Multiple-selection survey items are more complex still. These are represented as strings comprising comma-separated integers, whose values typically refer to non-ordinal categories. We coerce these to `list[Enum]` types, which have all the benefits of the categorical enums discussed above, and enable simpler programmatic analysis (e.g., testing for set membership, determining differences in response sets between waves).

Finally, we perform a series of value updates to ensure consistency between responses to different questions. This consists in replacing all empty string responses with `null` values[10], and re-mapping integer columns (both numeric and ordinal) such that the minimum value is zero.

9.2.2 Cleaning

Following schema normalisation, we then perform a cleaning stage, which primarily includes filtering out invalid or problematic data. We only filter out entire survey responses, rather than answers to specific survey items. Survey responses may be removed for any one of three reasons.

Firstly, we remove responses where the `participant_id` column is `null`, such that the individuals cannot be traced across survey waves. Secondly, we remove any responses from participants who fail the survey validity check in *any* of their participation waves. Thirdly, we remove any responses which fail either of the response-value or null-value validation checks described in the previous section (Chapter 9.1). We recognise that these issues are not necessarily problematic in every research context (e.g., cross-sectional studies are unconcerned with tracing an individual's responses across waves), and thus have made each condition optional in the dataset construction code.

9.2.3 Transformation

Finally, we apply several transformation steps with the goal of simplifying manipulation and analysis of the existing dataset, or enriching it with additional information.

The original dataset includes two survey items querying participants' alignment with the two largest US political parties (i.e., the Republicans and the Democrats). The first item asks whether the participant whether they consider themselves as *Republican*, *Democrat*, or *Independent*. Those who respond *Independent* are then presented a follow-up question asking whether they *lean* toward either (or neither) party. We replace these two survey items with a single variable combining the responses, with possible values:

$$\text{Democrat} \prec \text{Leaning Democrat} \prec \text{Independent} \prec \text{Leaning Republican} \prec \text{Republican}$$

[10] Note that this particular value mapping must happen *after* null-value validation (Chapter 9.1).

While this supplements the first item with additional information, we note two potential limitations of this transformation. Firstly, some participants who respond *Republican* or *Democrat* to the first question may more accurately describe themselves as only leaning in a given direction, if they had been aware of the (conditional) second question. Therefore the 'Leaning' responses are perhaps better interpreted as 'Independent with Republican/Democrat tendencies'.

Secondly, our implicit assumption that *Independent* is a midpoint between *Republican* and *Democrat* would misrepresent individuals who consider themselves 'right-of-Republican' or 'left-of-Democrat'. However, the available data does not allow us to distinguish such cases. Furthermore, this issue is also present in the original first question.

Our second transformation concerns survey questions with different variants, which are conditionally displayed based on treatment group membership. The original dataset includes, for each treatment group associated with a given question, a separate response column, as well as a binary indicator column describing group membership. Since group membership is always mutually exclusive in the dataset, we coalesce these various columns, replacing them with singular response and group membership columns. The group membership column contains integer indexes as opposed to the original binary indicators. In some cases, different treatment groups have an associated numerical value. For instance, the `ccSolveX` items query support policies compensating communities affected by climate change, with the treatment group specifying the cost (X) of such a policy to the participant. In these cases we supplement the dataset with an additional column containing the associated numerical value, simplifying downstream analyses.

Finally, we improve the dataset organisation by constructing a database comprising tables for survey item and question metadata and responses, as well as participants themselves (Figure 9.5). The requirement for such as database arose during the construction of the climate beliefs dataset used for model calibration (Chapter 9.3). The original data, comprising a codebook (a spreadsheet) and an RData file containing rows of participant responses, is not immediately amenible to our task, which requires reasoning about both survey logic and participation in tandem. The constructed database primarily serves to link survey metadata (extracted from the codebook) with survey response data.

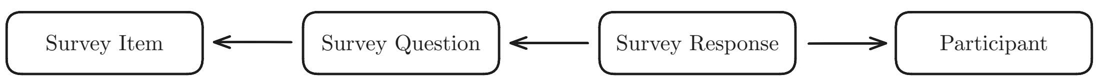


Figure 9.5: The survey database comprises tables for survey items (underlying concepts assessed in the survey), survey questions (the particular mode in which an item is assessed in a given wave or treatment condition), survey responses, and participants.

Recognising the hierarchical structure of the longitudinal dataset, in which a single survey item may vary in question text or response schema between waves or treatment conditions, we separate the concepts of *survey items* (the underlying concept being assessed) and *survey questions* (the particular mode of assessment). Survey questions are associated with particular waves, participant types (i.e., new or repeating), and treatment conditions, but may be related under a common survey item. *Survey responses* are associated with a particular question as well as a *survey participant*. We include a separate table for participants which records wave participation as well as the initial wave in which a participant has responded.

The database is stored on disk as a collection of parquet files (*Parquet*, n.d.), which preserve complex data type information and are supported in both R (e.g., using Arrow) and Python (e.g., using Pandas or Polars).

9.3 Climate beliefs dataset

In light of the complexities and breadth of content of the **placeholder dataset**, we construct a smaller, targeted dataset of beliefs and attitudes relating to climate change, which we expect – on theoretical grounds – to exhibit interdependent behaviour. We will refer to this targeted dataset as the **climate beliefs dataset**. The primary motivator for constructing this dataset is the calibration of the non-equilibrium belief system model (Chapter 4.3).

We aim for 7–10 variables in the final set, so as to keep both the number of model parameters and parameter uncertainty sufficiently small. We first filter the survey items to consider only those which assess cognitive states (e.g., beliefs, attitudes, opinions, stances). We further restrict our attention to items which either relate directly to climate change, or which are expected to influence climate-related cognitive states (e.g., political alignment).

Within this restricted set of survey items our intention is to select a collection of variables which: (i) exhibit related behaviour, (ii) maximise the number of observations, and (iii) satisfy the requirements of the parameter estimation method used for model calibration. We will discuss the latter two requirements now, and return to the first shortly. We require at least two waves of responses from each participant, as the model is defined by a time-dependent Markov process, and thus parameter estimation involves fitting a conditional probability distribution where each belief state depends on the previous one. Moreover, each participant's responses should be equispaced, with spacing being consistent across individuals, and the dataset must contain no missing values. In addition to these requirements, we prioritise including several variables of interest:

1. **CC Worry**: Concern about current and future climate change
2. **CC Others Worry**: Belief regarding *others'* level of worry about climate change
3. **CC Human**: Belief that climate change is caused by human activities

The first is generally considered influential in shifting related attitudes, while the second is expected to be influential on the first, as a perceived descriptive norm. Finally, while most inividuals in the USA believe in the *existence* of climate change, there is relatively greater variation in beliefs about it's *causes*. Hence the third item is an interesting target for intervention studies. (**TODO: Add citations**)

Given the above requirements and the objective to maximise the number of observations, we identify Waves 3 & 4 as suitable candidates. These waves include 1693 repeat observations (excluding survey errors as described in the previous subsection). After removing items with no substantial correlations, and small groups of items which exhibit only internal correlations, we identify 17 relevant survey items, comprising both beliefs (epistemic positions, Table 9.2) and attitudes (qualitative evaluations, Table 9.3).

To reduce the set of candidate variables to the target range of 7–10, we proceed to identify groups of similar or redundant variables which may be removed or combined into interpretable index variables. We use two primary measures to investigate similarity: pairwise partial correlation, and vector autoregression (Epskamp, van Borkulo, et al., 2018; Epskamp, Waldorp, et al., 2018).

Partial correlation measures the correlation between a pair of variables X and Y after conditioning on all remaining variables Z:

$$\rho(X, Y) = \mathrm{Corr}(X, Y \mid Z) \tag{9.2}$$

In particular, $|\rho(X, Y)|$ is non-zero if X and Y exhibit *linear* correlation which is not shared with the remaining variables in Z. For instance, consider the fork structure $X \leftarrow A \rightarrow Y$ where X and Y are probabilistic linear functions of A. If the value of A is unknown, then X and Y are dependent. However, once A is known, the dependency structure is broken. Thus $\mathrm{Corr}(X, Y)$ is nonzero, while $\rho(X, Y) = 0$.

Figure 9.6 displays the pairwise partial correlation between each pair of variables in the filtered set. Note that we have coded all variables such that the partial correlation matrix entries are predominantly positive. Cells with magnitude below 0.05 have been masked for visual clarity. Rows and columns are ordered using hierarchical clustering.

Item	Question text	Response schema
CC Real	Do you think that climate change is happening?	• Yes • No • Don't know
CC Human (*)	Do you think rising temperatures are a result of human activities, natural causes, or both?	• Not happening • Natural causes • Human activities • Both
CC Impact (world)	How much do you think climate change is currently harming the world in general?	• Not at all • Only a little • A moderate amount • A great deal
CC Impact (wealthy)	How much do you think climate change is currently harming wealthy communities within the United States?	• Not at all • Only a little • A moderate amount • A great deal
CC Impact (poor)	How much do you think climate change is currently harming poor communities within the United States?	• Not at all • Only a little • A moderate amount • A great deal
CC Impact (comm)	How much do you think climate change is currently harming your local community?	• Not at all • Only a little • A moderate amount • A great deal
CC Others Worry	How worried do you think most Americans are about global warming/climate change these days?	• Not at all worried • Not very worried • Somewhat worried • Very worried

Table 9.2: * Re-mapped to: includes/does not include human activities

Partial correlations are limited, however, in that they capture only instantaneous (or 'contemporaneous') relations between variables, while the non-equilibrium belief system model is concerned with temporal relationships. Therefore we also examine the temporal and contemporaneous networks obtained using vector autoregression (VAR) with a $\delta t = 1$ time lag (Brandt & Williams, 2007). These networks are derived from the solution to the following regression problem, described in Epskamp, Waldorp, et al. (2018):

$$\begin{aligned} \boldsymbol{y}^t &= \boldsymbol{B}\boldsymbol{y}^{t-1} + \varepsilon^t \\ \varepsilon^t &\sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Theta}) \end{aligned} \tag{9.3}$$

Where $\boldsymbol{y}^t$ is the vector of observed variable values for each individual at time t. The *temporal network* matrix $\boldsymbol{B}$ comprises linear next-state predictors, while the *contemporaneous network* matrix $\boldsymbol{K} = \Theta^{-1}$ describes the pairwise conditional correlations

Item	Question text	Response schema
CC Worry	How worried are you about current and future global warming/climate change?	• Not at all worried • Not very worried • Somewhat worried • Very worried
Weather Worry	How worried are you about an extreme weather event or natural disaster happening to you personally in the next year?	• Not at all • Only a little • A moderate amount • A great deal
CC Responsibility	It is important that individuals take action on issues of climate change.	5-point Likert agreement scale
CC Scientists	Scientists with appropriate expertise should guide how we respond to climate change	5-point Likert agreement scale
Policy: ICA (*)	Do you favour an international agreement committing the USA and other countries to reduce their carbon emissions?	• Yes and legally binding) • Yes but *not* legally binding) • No
Policy: Tax fuel	How much do you support/oppose a tax on the production/distribution of carbon-based fuelds?	5-point Likert support scale
Policy: Auto	How much do you support/oppose stronger carbon emissions standard for car manufacturers?	5-point Likert support scale
Policy: Env. Reg.	Which statement comes closer to your views?	• Stricter environmental regulations cost too many jobs and hurt the economy • Stricter environmental regulations are worth the cost
Political affiliation (**)	In politics today, do you consider yourself a:	• Republican • Leaning Republican • Independent • Leaning Democrat • Democrat
Political ideology	What is your political ideology?	• Very conservative • Conservative • Moderate • Liberal • Very liberal

Table 9.3: * Re-mapped to: Yes/No

** Constructed from separate items for identification and leaning

which remain after accounting for temporal effects. The entries of $\boldsymbol{K}$ are analogous to

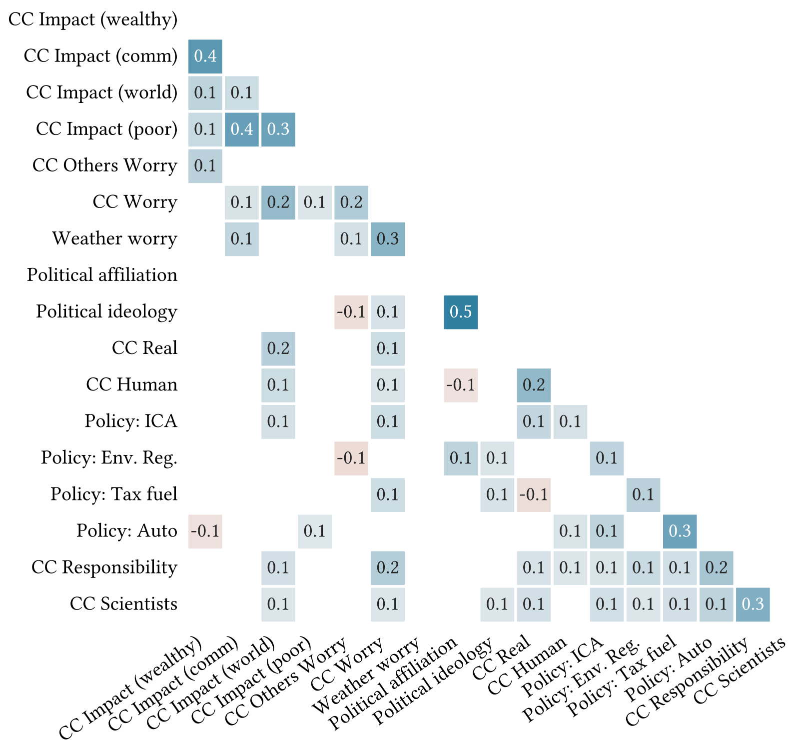


Figure 9.6: Pairwise partial correlation for the filtered candidate set (Table 9.2, Table 9.3). Cells with $|\rho(X,Y)| < 0.05$ are masked for visual clarity. Rows and columns are ordered using hierarchical clustering over the distance matrix: $d_\rho(X,Y) = 1 - |\rho(X,Y)|$.

partial correlations, constituting the conditional correlation between X and Y given the remaining variables.

Figure 9.7 shows the contemporaneous and temporal network matrices for the filtered set of candidate variables. The contemporaneous matrix is symmetric, thus the upper triangle is not shown.

We observe some apparent commonalities between the partial correlation and VAR networks. First, the *CC Impact* variables exhibit relatively strong internal relations. External relations with variables outside this set are typically weaker, with the exception of *CC Impact (world)*, which has several non-trivial partial correlations with other variables, including other climate-related beliefs. All variables in this group, except *CC Impact (wealthy)*, also have non-trivial partial correlations with *CC Worry* — one of our variables-of-interest. Second, the policy variables, as well as *CC Responsibility* and *CC Scientists*, also exhibit substantial connectivity in the partial correlation network. This is diminished in the contemporaneous VAR network, but visible in the clustering of the temporal network. The similarities among these variables in the rows and columns of the temporal matrix indicate that they fill similar roles as predictors, and are also

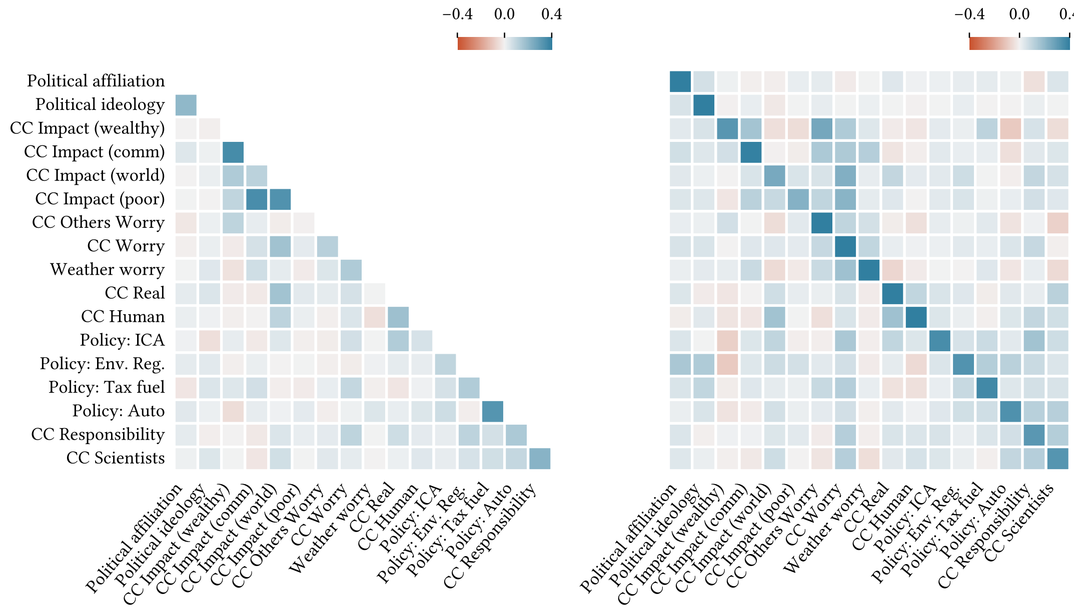


Figure 9.7: Contemporaneous (left) and temporal (right) network matrices for the filtered candidate set (Table 9.2 Table 9.3). Entries in row i of the temporal matrix are the regression predictors for the subsequent value of variable X_i. Rows and columns are ordered using hierarchical clustering on the distance matrix: $d_{\boldsymbol{K}}(X,Y) = 1 - |\boldsymbol{K}|$.

predicted similarly. Thirdly, *Political affiliation* and *Political ideology* are highly related in the partial correlation matrix, and also exhibit similar behaviour in the temporal network.

Based on these observations, we construct three index variables: (i) *CC Impact*, comprising the four climate impact beliefs, (ii) *Politics*, comprising the measures of political affiliation and ideology, and (iii) *CC Action*, comprising the four policy variables, as well as *CC Responsibility* and *CC Scientists*.

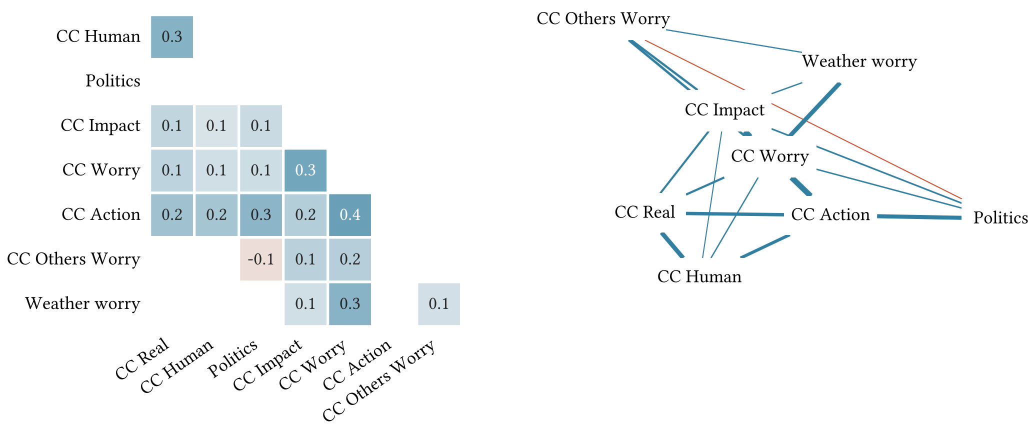


Figure 9.8: **TODO**

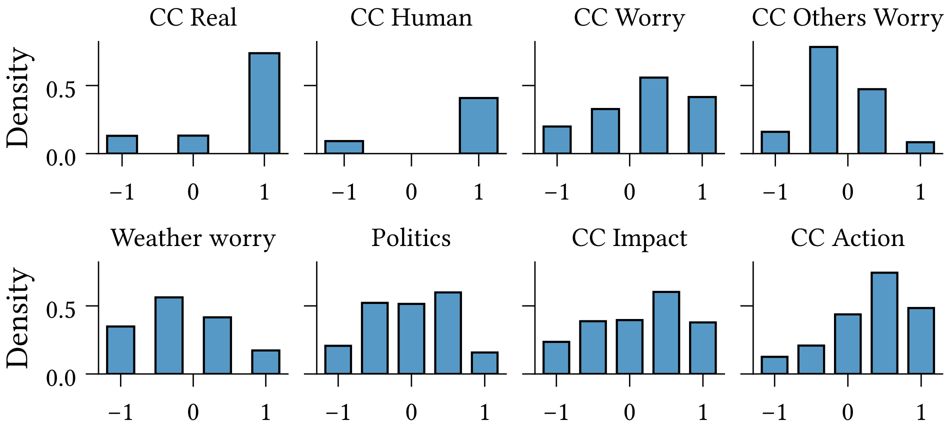


Figure 9.9: **TODO**

Chapter 10

Related work

10.1 Plan

1. Psychological/philosophical theories of beliefs and attitudes
   - Cognitive consistency:
     - Heider (balance theory) (Heider, 1946), Insko & Schopler (triadic consistency) (Insko & Schopler, 1967)
     - Festinger (cognitive dissonance) (Festinger, 1962)
   - Fishbein (Fishbein, 1966)
   - Quine (Quine, Willard van Orman & Ullian, Joseph Silbert, 1978)

2. Belief system modelling:

   - Schools of thought:
     - Triadic consistency (Rodriguez et al., 2016)
     - Parallel constraint satisfaction (Monroe & Read, 2008)
     - Bayesian networks (Cook & Lewandowsky, 2016; Powell et al., 2023)
     - Structural equation modelling?
     - Regularised partial correlation networks
     - Ising
       - CAN (Dalege et al., 2016)
       - Social influence: Network of belief (Dalege et al., 2025), hierarchical Ising model (van der Maas et al., 2020)

   - What theory/assumptions does each model draw on?

   - Difference between Latent variable and empirical network models (Dalege et al., 2016)

- Case studies — how have the models been applied?
  - Theoretical explanations for known phenomenon
  - Analysis of survey data

3. Interventions
   - Modelling, testing effects (Powell et al., 2023)
   - Theoretical (Dalege et al., 2025)
   - In-practice (The Workshop, 2024)

4. Inference
   - Between-individual limitations (Brandt & Morgan, 2022)

Chapter 11

Conclusions and future work

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Chapter 12

Ethics and Data Management

A new requirement for the thesis is that there must be a short section in which you reflect on the ethical aspects of your project. This requirement is related to one of the final objectives that a graduated student of the Master of Computational Science must meet: “The graduate of the program has insight into the social significance of Computational Science and the responsibilities of experts in this field within science and in society“. You don’t need to devote an entire chapter to this; a short section or paragraph is sufficient.

I acknowledge that the thesis adheres to the ethical code (https://student.uva.nl/en/topics/ethics-in-research) and research data management policies (https://rdm.uva.nl/en) of UvA and IvI.

The following table lists the data used in this thesis (including source codes). I confirm that the list is complete and the listed data are sufficient to reproduce the results of the thesis. If a prohibitive non-disclosure agreement is in effect at the time of submission “NDA” is written under “Availability” and “License” for the concerned data items.

Bibliography

Bantilan, N., Markey, N., Jean-Francois Zinque, Matache, C., Hahmann, F., Hart, G., Albertazzi, R., Mackesey, S., Loukianov, A., Matthew, Baumgartner, M., Radojković, N., Craigie, R., Chr1st1ank, Fleimgruber, Tfwillems, Weeden, A., Singh, A., Ripshtos, A., … Myatt, J. (2022, August). *Unionai-Oss/Pandera: Beta Release v0.12.0b0*. https://doi.org/10.5281/ZENODO.3385265

Barber, R. F., & Drton, M. (2015). High-Dimensional Ising Model Selection with Bayesian Information Criteria. *Electronic Journal of Statistics*, *9*(1), 567–607. https://doi.org/10.1214/15-EJS1012

Brandt, M. J., & Morgan, G. S. (2022). Between-Person Methods Provide Limited Insight about within-Person Belief Systems. *Journal of Personality and Social Psychology*, *123*(3), 621–635. https://doi.org/10.1037/pspp0000404

Brandt, P. T., & Williams, J. T. (2007). *Multiple Time Series Models*. SAGE.

Bumann, S. (2021). What Are the Determinants of Public Support for Climate Policies? A Review of the Empirical Literature. *Review of Economics*, *72*(3), 213–228. https://doi.org/10.1515/roe-2021-0046

Cardy, J. (1996). *Scaling and Renormalization in Statistical Physics*. Cambridge University Press.

Chen, J., & Chen, Z. (2008). Extended Bayesian Information Criteria for Model Selection with Large Model Spaces. *Biometrika*, *95*(3), 759–771.

Christensen, K., & Moloney, N. R. (2005). *Complexity and Criticality*. Imperial College Press.

Cook, J., & Lewandowsky, S. (2016). Rational Irrationality: Modeling Climate Change Belief Polarization Using Bayesian Networks. *Topics in Cognitive Science*, *8*(1), 160–179. https://doi.org/10.1111/tops.12186

Dalege, J., Borsboom, D., Van Harreveld, F., Van Den Berg, H., Conner, M., & Van Der Maas, H. L. J. (2016). Toward a Formalized Account of Attitudes: The Causal Attitude Network (CAN) Model. *Psychological Review*, *123*(1), 2–22. https://doi.org/10.1037/a0039802

Dalege, J., Galesic, M., & Olsson, H. (2025). Networks of Beliefs: An Integrative Theory of Individual- and Social-Level Belief Dynamics. *Psychological Review*, *132*(2), 253–290. https://doi.org/10.1037/rev0000494

Egan, P. J. (2020). Identity as Dependent Variable: How Americans Shift Their Identities to Align with Their Politics. *American Journal of Political Science*, *64*(3), 699–716. https://doi.org/10.1111/ajps.12496

Epskamp, Borsboom, et al. (2018). Estimating Psychological Networks and Their Accuracy: A Tutorial Paper. *Behavior Research Methods*, *50*(1), 195–212. https://doi.org/10.3758/s13428-017-0862-1

Epskamp, van Borkulo, et al. (2018). Personalized Network Modeling in Psychopathology: The Importance of Contemporaneous and Temporal Connections. *Clinical Psychological Science*, *6*(3), 416–427. https://doi.org/10.1177/2167702617744325

Epskamp, Waldorp, et al. (2018). The Gaussian Graphical Model in Cross-Sectional and Time-Series Data. *Multivariate Behavioral Research*, *53*(4), 453–480. https://doi.org/10.1080/00273171.2018.1454823

Festinger, L. (1962). Cognitive Dissonance. *Scientific American*, *207*(4), 93–106.

Fishbein, M. (1966). The Relationships between Beliefs, Attitudes, and Behavior. In *Cognitive Consistency: Cognitive Consistency* (pp. 199–223). Elsevier.

Foygel, R., & Drton, M. (2010). Extended Bayesian Information Criteria for Gaussian Graphical Models. *Advances in Neural Information Processing Systems*, *23*.

Heider, F. (1946). Attitudes and Cognitive Organization. *The Journal of Psychology*, *21*(1), 107–112. https://doi.org/10.1080/00223980.1946.9917275

Holshue, M. L., DeBolt, C., Lindquist, S., Lofy, K. H., Wiesman, J., Bruce, H., Spitters, C., Ericson, K., Wilkerson, S., Tural, A., Diaz, G., Cohn, A., Fox, L., Patel, A., Gerber, S. I., Kim, L., Tong, S., Lu, X., Lindstrom, S., … Pillai, S. K. (2020). First Case of 2019 Novel Coronavirus in the United States. *New England Journal of Medicine*, *382*(10), 929–936. https://doi.org/10.1056/NEJMoa2001191

Huddy, L., Mason, L., & Aarøe, L. (2015). Expressive Partisanship: Campaign Involvement, Political Emotion, and Partisan Identity. *American Political Science Review*, *109*(1), 1–17. https://doi.org/10.1017/S0003055414000604

Insko, C., & Schopler, J. (1967). Triadic Consistency. *Psychological Review*, *74*(5), 361–376.

Iyengar, S., Sood, G., & Lelkes, Y. (2012). Affect, Not Ideology: A Social Identity Perspective on Polarization. *Public Opinion Quarterly*, *76*(3), 405–431. https://doi.org/10.1093/poq/nfs038

Lee, E. D., Broedersz, C. P., & Bialek, W. (2015). Statistical Mechanics of the US Supreme Court. *Journal of Statistical Physics*, *160*(2), 275–301. https://doi.org/10.1007/s10955-015-1253-6

Lee, S., Vu, H. T., Thaker, J., Verner, M., Goldberg, M. H., Carman, J., Rosenthal, S. A., & Leiserowitz, A. (2025). Variations in Climate Change Belief Systems across 110 Geographic Areas. *Nature Climate Change*, *15*(9), 947–953. https://doi.org/10.1038/s41558-025-02410-1

Mason, L. (2018). Ideologues without Issues: The Polarizing Consequences of Ideological Identities. *Public Opinion Quarterly*, *82*(S1), 866–887. https://doi.org/10.1093/poq/nfy005

Monroe, B., & Read, S. (2008). A General Connectionist Model of Attitude Structure and Change. *Psychological Review*, *115*(3), 733–759. https://doi.org/10.1037/0033-295X.115.3.733

Nguyen, H. C., Zecchina, R., & Berg, J. (2017). Inverse Statistical Problems: From the Inverse Ising Problem to Data Science. *Advances in Physics*, *66*(3), 197–261. https://doi.org/10.1080/00018732.2017.1341604

Nocedal, J., & Wright, S. J. (2006). *Numerical Optimization*. Springer.

Parquet. (n.d.).

Powell, D., Weisman, K., & Markman, E. M. (2023). Modeling and Leveraging Intuitive Theories to Improve Vaccine Attitudes. *Journal of Experimental Psychology: General*, *152*(5), 1379–1395. https://doi.org/10.1037/xge0001324

Quine, Willard van Orman, & Ullian, Joseph Silbert. (1978). *The Web of Belief* (Vol. 2). New York: Random House.

Rodriguez, N., Bollen, J., & Ahn, Y.-Y. (2016). Collective Dynamics of Belief Evolution under Cognitive Coherence and Social Conformity. *PLOS ONE*, *11*(11), e0165910. https://doi.org/10.1371/journal.pone.0165910

Roser-Renouf, C., Maibach, E. W., Leiserowitz, A., & Zhao, X. (2014). The Genesis of Climate Change Activism: From Key Beliefs to Political Action. *Climatic Change*, *125*(2), 163–178. https://doi.org/10.1007/s10584-014-1173-5

Shao, W., & Hao, F. (2020). Approval of Political Leaders Can Slant Evaluation of Political Issues: Evidence from Public Concern for Climate Change in the USA. *Climatic Change*, *158*(2), 201–212. https://doi.org/10.1007/s10584-019-02594-4

Tibshirani, R. (1996). Regression Shrinkage and Selection Via the Lasso. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, *58*(1), 267–288. https://doi.org/10.1111/j.2517-6161.1996.tb02080.x

Unsworth, K. L., & Fielding, K. S. (2014). It's Political: How the Salience of One's Political Identity Changes Climate Change Beliefs and Policy Support. *Global Environmental Change*, *27*, 131–137. https://doi.org/10.1016/j.gloenvcha.2014.05.002

van der Maas, H. L. J., Dalege, J., & Waldorp, L. (2020). The Polarization within and across Individuals: The Hierarchical Ising Opinion Model. *Journal of Complex Networks*, *8*(2), cnaa010. https://doi.org/10.1093/comnet/cnaa010

van Noord, J., Turner-Zwinkels, F. M., Kesberg, R., Brandt, M. J., Easterbrook, M. J., Kuppens, T., & Spruyt, B. (2025). The Nature and Structure of European Belief Systems: Exploring the Varieties of Belief Systems across 23 European Countries. *European Sociological Review*, *41*(1), 143–161. https://doi.org/10.1093/esr/jcae011

Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., Van Der Walt, S. J., Brett, M., Wilson, J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., … Vázquez-Baeza, Y. (2020). SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nature Methods*, *17*(3), 261–272. https://doi.org/10.1038/s41592-019-0686-2

The Workshop. (2024). *How to Talk about Community Planning for Climate Disruption* [Technical report].

Ziegler, A. (2017). Political Orientation, Environmental Values, and Climate Change Beliefs and Attitudes: An Empirical Cross Country Analysis. *Energy Economics*, *63*, 144–153. https://doi.org/10.1016/j.eneco.2017.01.022

Appendix A

Derivations

A.1 Parameter estimation

A.1.1 Log-likelihood

Equation 4.5

A.1.2 Expected log-likelihood

Equation 4.6

A.1.3 Partial derivatives of expected log-likelihood

A.2 Dataset

A.2.1 Binarisation probability

Equation 4.1

Appendix B

Additional results

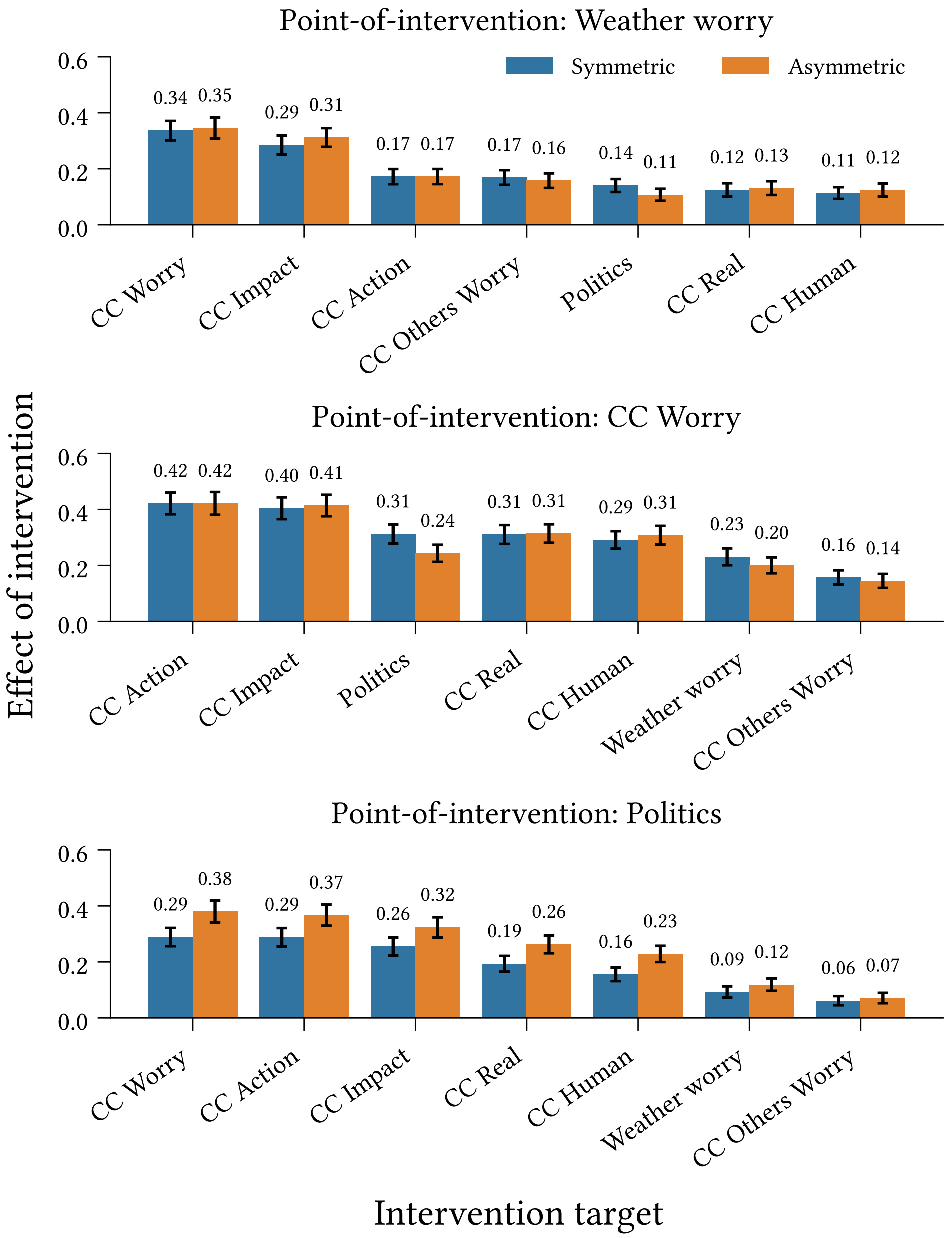

Figure B.1: Outbound effect of intervention (Definition 6.2.1) for interventions targeting `Weather Worry`, `CC Worry`, and `Politics`, in symmetric and asymmetric belief system models calibrated to the climate beliefs dataset. Measurements taken at $t = 10$ (approximately five years in the models' timescale). Intervention strength: $\delta_h = 2.5$. Confidence intervals display 1.96 standard deviations around the mean effect, measured across 500 repeated simulations.